

UNIVERSITY PARTNER



Project and Professionalism (6CS007)

FYP Final Report

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3. Abstract

The title for my final year project is making a web-based e-learning platform for barista training. E-learning can be referred as a form of studying outside of the classroom with the use of computer. The adoption of education via technology has been effective in learning skills acquiring knowledge. (The Economic Times, 2021) The project is created using MERN stack. Firstly, the backend API were tested using Postman and further development was done of the front end using React.

5. Introduction

5.1 Project Briefing

The name of this e-learning system is Kaldiable meaning “Kaldi” is the name of the person who first discovered coffee and “able” means being able to learn and make it happen. This system will be designed based on all the knowledge that a Barista should learn and make them well prepared for this field. The web application is developed using full stack developing i.e. MERN stack. Barista training program is to teach students how to grind coffee beans and blend it to make various coffee-based drinks as well as other types of drinks. Cafés are place where people usually like to hangout and prefer to go for work meetings due to its ambience, aroma of coffee all around the place. So, this business is growing daily with high demands of skilled barista staffs. Barista training in Nepal costs about 15,000 to 30,000 rupees which lasts for 2 weeks with internship (Barista Training Nepal, 2021), \$200 to \$1,249 in the US (coursetakers, 2021), 60 to 180 euros in the European countries (EUROPEAN BARTENDER SCHOOL, 2021), and about \$1,234.00 in China (TORCH COFFEE COMPANY , 2021). Therefore, the cumulative cost is expensive just to pursue to learn art of coffee making.

Furthermore, Barista art is also very much in practice for the students seeking abroad for further studies. They learn this course with a certification that allows them to work a part time job as a Barista while studying. This field has provided many people with employment as a Barista trainer,

or a Barista. The major hurdle in this field is that the course is costly which cause many to leave their dream of pursuing a career in coffee art. Therefore, in my final year project, I will be developing an e-commerce website for barista learning that facilitates this course online and provide the certification on the successful completion of the course.

The unemployment rate in 2021 in Nepal is expected to reach 3.60 percent (TRADING ECONOMICS, 2021). This rate of unemployment is due to the lack of opportunities. This system will be designed to create opportunity to those who are seeking to learn some skills and earn a living being a Barista. The main issue that is reflected to learn this course is that practically it is quite expensive and everyone cannot afford it. This system will make this course available for free so that interested individual can learn skills and get certification for their successful completion. Most people seeking jobs and studies abroad will also be benefitted by this system. The advantage of learning barista skills is it does not require your qualification in studies it helps you earn a job based on the skills learnt in short period of time. This website will help them on learning the skills necessary to qualify as a barista.

5.1.1 Problem Domain

The main concept behind giving it a reality picture is due to the expensiveness of learning coffee making professionally. Due to this reason many people do not get to pursue their interest in this field. Barista training is mostly renowned among the students flying abroad for further studies it has benefitted them as a part time job while their stay. It is an employment source for many and since coffee is a booster for starting a day for many working people it is a very growing business these days. As per the recent COVID outbreak, people were compelled to do study, work adopting online platform. Likewise, providing training online can help people to think critically and self-learn. Barista demand in the market prevailing and giving employment opportunities to lot of people so provided training based on this system is to create a skillful Barista.

5.2 Aims

- how html can give the structure for the website
- how CSS beautifies and makes that structure alive
- how JavaScript adds behavior to that beautiful structure
- how a single page application is implemented just by routing in React
- how Node js as a backend works
- how online e-learning can be implemented to teach a course
- how database works for the storage purpose
- how a website can be made responsive
- how a project development process is planned with dates in a chart and completing work based on that chart
- how a diagram is formed to picture the functionality of the whole project
- how the testing process is implemented
- time management during development until the deployment of the project

5.3 Objectives

The goal of this project is to develop an e-learning system for making and serving perfect cup of drinks specially related to coffee. This system will be used for developing skilled baristas in the business sector providing employment and experience.

To achieve the goal, mainly the following objectives will be focused which are briefly described below:

1. Develop a user registration system that allows the user to register before continuing further learning so that their progress gets saved and they can continue with their learning from where they left whenever they feel like after just signing in.
2. Develop admin registration system that can only be seen by the administrators of the system and enable the admins to make changes in the system accordingly.

3. Develop a user interface (UI) for the learners. This UI will be web-based and user friendly. It allows the users to check all the important contents and skills that they will need to know about being a barista.
4. Develop a UI for administrators. This UI will be Node js template where the administrator will be allowed to upload and delete videos, recipes, images, etc. Moreover allow certificates to the users after successfully completing the course.
5. Develop a UI for proper certification that the users will be able to download after completion of the course.

5.4 Artefact

The system is used to achieve knowledge and learn skills in the field of barista. It is an online based system that can provide a user with a certificate after successful completion of the course. It consists of history of coffee origin, with all the images of the tools required for learning with the description of where to use it. Even videos are provided to make the learning effective. Tips and tricks are also mentioned to master work of Barista. There is also a quiz game to test the knowledge level of the users to revise everything that have so far learned from the course.

5.4.1 FDD

The functional decomposition diagram shows the functions that each sub systems will carry out in the proposed system. The admin panel allows the admin to create another admin and also delete them and the admins can disable or enable certificate to the users. The user panel allows the user to view the course offerings and provide them with enroll option after which they can participate and take quiz and claim certificate on completing the course successfully. The panel for course, quiz and video will have create, read, update, and delete functionality will be handled by the admins.

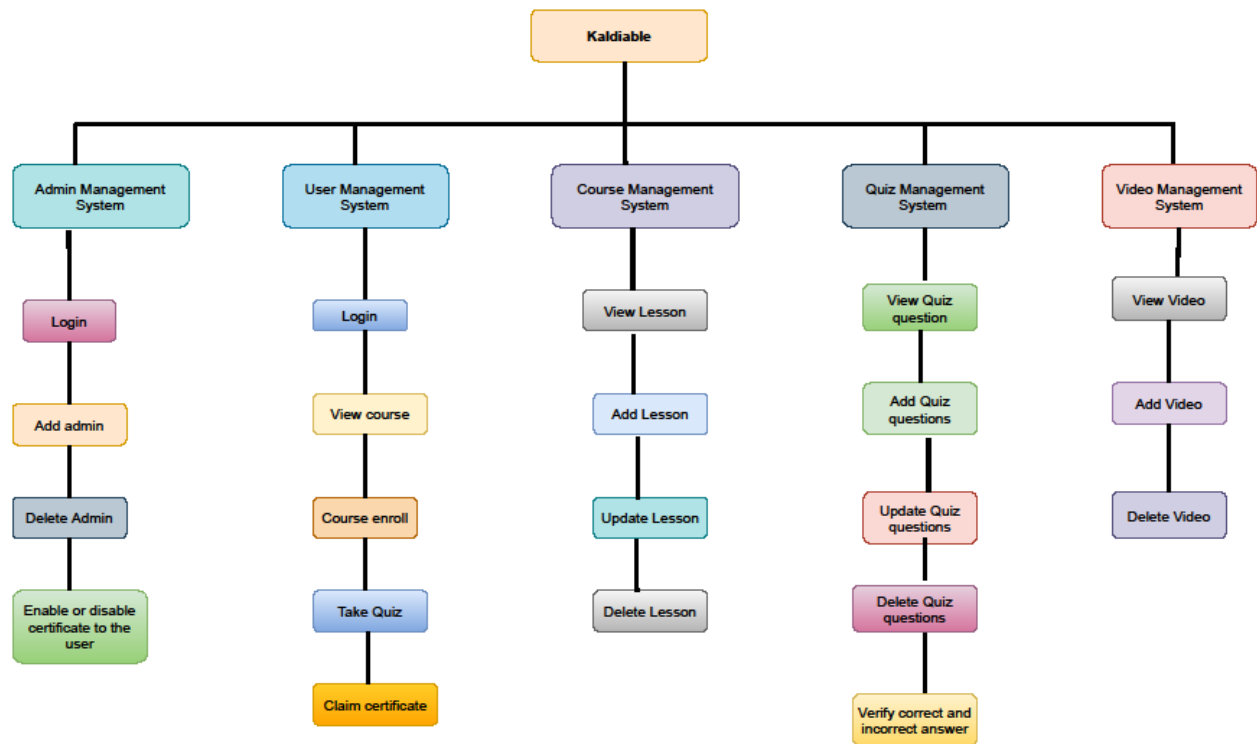


Figure 1: Functional Decomposition Diagram (FDD)

5.4.2 WBS

WBS ID	Task	Fill Color	Text Color
0	Kaldiabale		
1	Title Research		
1.1	System decided		
1.2	Technology and tools selection		
1.3	Project name selection		
2	Project proposal		
2.1	Aims		
2.2	Objectives		
2.3	Artefact		
2.3.1	Product backlog		
2.3.2	UML diagrams		
2.3.3	Erd diagrams		
2.3.4	Wireframes		
2.3.5	Literature Review		
3	Development		
3.1	Front end development		
3.2	Back end development		
4	Test-Plan Schedule		
4.1	Gantt chart		
5	Professionalism Report		
5.1	Advanced Artefact		
5.2	Literature Review		
6	Final analysis of the whole system		
6.1	Wireframes		
6.2	Design analysis		
6.3	Code analysis		
6.4	Front end analysis		
6.5	Back end analysis		
7	Presentation slides		
7.1	Presentation		
7.2	System Kick-off		

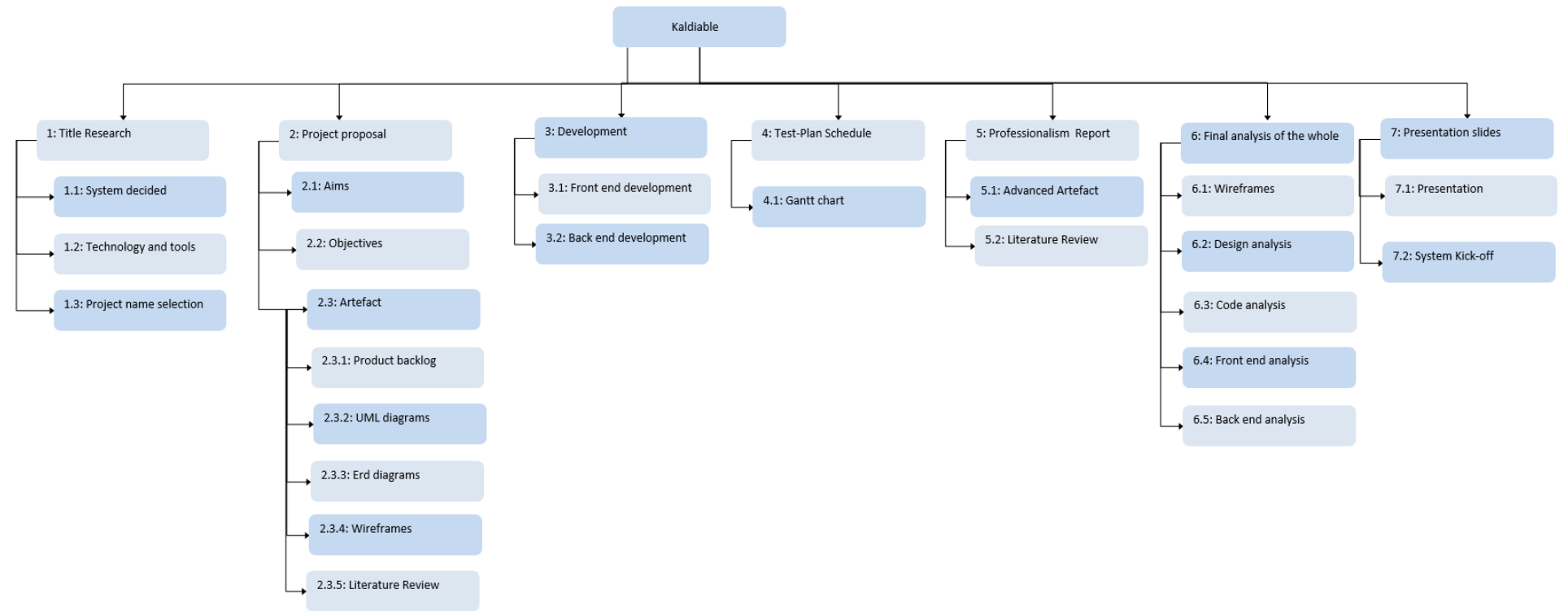


Figure 2: Work breakdown system

5.5 Academic Questions

How can it contribute and help the employment sector? The name of my e-learning system would be Kaldiable meaning “Kaldi” is the name of the person who first discovered coffee and “able” means being able to learn and make it happen. This system will be designed based on all the knowledge that a Barista should learn and make them well prepared for this field.

- What opportunities can the project provide?
 - Employment opportunities to people who wants to earn through skills from making beverages. Also students flying abroad can get benefitted by this system to carry part time job along with their studies.
- What possibilities can the project provide from a learner’s perspective?
 - Learners develop skills based on their capability and develop the attitude of practicing and learning on their own through provided resources.
- What contents will be displayed in the website for the enhancement of better learning?
 - Visuals and detailed information with practice can enhance and upgrade the traditional form of learning with teachers and students in a constricted area. It develops better understanding with step by step learning pattern on individual level.
- How e-learning platforms enhance critical thinking and develop self-learning ability?
 - E-learning platform is based on accessing a system and based on the resources provided on that system the learners need to grasp and benefit themselves to acquire knowledge to achieve their wanted goals i.e. to achieve a recognition that the person is skilled and eligible after taking the course.

5.6 Scope and Limitations

5.6.1 Scopes

- Cost and time effective – Pursuing studies online saves times as a student does not have to visit somewhere to learn a course just by staying at a place he/she can access the course offerings with a click. Furthermore, it saves money from giving resources needed for the course by making it freely available on the system online.
- Accessible – E-learning can be accessed at any time when a learners prefers to as well as it covers every topic that we are supposed to know about that provided course. Lastly, on completion of the course it motivates a learner by allowing them to download their course completion certificate with their name printed on it. This feature provided by e-learning thus as a learner encourages them to pursue further learning.
- Online classroom environment – Every learners do not prefer traditional learning system. Also, every student does not have same learning capabilities some are slow at catching some are fast these hurdles are addressed by online system. Learners can be assured that they will grasp knowledge in every module they access in the system with their own comfort level of learning. (Raju, et al., 2019)

5.6.2 Limitations

- The limitations on of using Kaldiable is the system stores data using MongoDB Atlas and without internet it cannot store data thus the system will not be able to function properly.
- The instructor needs to prepare all the resources for every modules and asks for their extra time and effort.
- Users might get instant problem or confusion during the accessing the system and the instructors will not be able to address their queries at that time and the process of learning gets slow due to late response from the tutor.
- Since it is an online learning platform there it may lead to piracy and plagiarism and there will not be any plagiarism caused by the users. (Thanji & Vasantha, 2018)

6. Literature Review

Integrating learning styles and adaptive e-learning system: Current developments, problems and opportunities

The learning ways preferred by students differs from one another some prefers visual based learning, some prefers group discussion based learning whereas some understands the topic through text, theories, practical experiments, etc. The level of understanding a topic for each students are different and the measurement of this difference provides an idea to develop means to monitor the learning patterns of students. And that developed means can in return help the teachers to get useful feedbacks to create a better learning environment for the students considering the majority learning paths. According to the various theories provided which suggests learning styles are fixed depending on the institution, learning strategies changes as per the situation.

The problems faced during the measurement of preference of students learning ability before were very biased as the student needed to evaluate their personal learning behavior. Likewise, the surveys conducted this way yielded more than 40 questions which is not feasible to update.

Furthermore the problems and the drawbacks helped in creation of an integrated e-learning system that helps in keeping tabs on the capability of students during and after the learning process through algorithms and data mining. Such features by e-learning system makes analyzing of big dataset easier to make preferred teaching techniques as per the online recorded behaviors of the students. The analysis made per the system usage did help in considering development of other adaptive framework for better learning techniques. The recent studies on promoting effective learning techniques through learning models have shown positive development in figuring what the students prefer on the basis of their study background.

The reviews on this research paper taken from various article mediums such as: - Google Scholar, science direct and even reviewing an implemented system related to computerized learning. This research paper focuses on integrating adaptive learning system based on the theories and implementation of that theories in applications that was reviewed from 51 papers. Among them 39 were journals and 12 were conference papers.

The development insights mentioned in this paper can be helpful for the development process of e-learning system as this paper mentions about the prediction of learning styles and how that is implemented in a system. It helps to understand the process on what a system should follow before reaching the development phase which were determining models that recorded learning attributes and based on that attributes algorithms should be selected for classification and after the evaluation of appropriate classification model a well adaptive system can be prepared.

The approach and steps followed to develop a well-integrated system with the proposed theories, selected attributes for preferred learning styles, classification algorithms selected for learning styles which on further research would have brought light to current ongoing problems and decide solutions for better development of learning platforms.

Among the number of learning theories mentioned in this paper Felder-Silverman's model stood out as the theory covered main learning styles and combined the nature of perception of students for pursuing methods of learning. The learning predictors can act as an medium to identify the main attributes necessary to determine learning preferences which are the log files as it keeps tabs on the frequents visits, time spent, participation in quiz, puzzles, downloaded materials and history of users that holds information such as: - gender, ethnicity which are not that important attributes but can help determine the learning styles and lastly personal source that are related with intelligence, knowledge, capacity to remember can be a factor to make an adaptive learning system.

The algorithms used for the classification in order to determine the adaptive learning system was rules-based which was based on statistical graph on various theories. Bayesian network was secondly most used algorithm which use probability approach. From the 51 studies most of the papers used single algorithm for classification whereas some used combination of two or more algorithms for classification. The interesting combination of algorithms was Naïve Bayes and decision tree that helped better in classifying visual and auditory and aside from this algorithm hidden Markov Chain was also applied which was better to monitor the sequential actions of students who studied line by line and students who jumps from sections to sections according to their preference. The newly developed application promoted educational games as studies showed that the students showed interest in learning through games made for educational perspective and came handy for student who had creativity score of above average.

The theory proposed by Felder-Silverman's was further analyzed to evaluate the students' learning outcomes. As mentioned before there are three main sources to determine learning styles which were determined by log files, static information and personalized sources. These findings and research can help as well as act as a baseline for researchers to develop an effective classification models and prediction models for adaptive learning system. The number of algorithms applied for the prediction had their own pros and cons. The most accepted ones were Bayesian network and rules-based as it showed the most promising results. The rules-based gives an ability to change the learning techniques if it needs any update over time. On the other hand Bayesian network does not leave out any attributes that are necessary to estimate the learning styles. These both algorithm are highly based on assumptions. Due to its dependency on assumptions it is hard to be transferred. So, the hybrid and ensemble methods were formed which yielded many benefits such as: - ensemble method reduces variance and overcomes the biasness of individual models whereas hybrid improves the accuracy by tuning and filtering the data.

In conclusion, none of the above mentioned theories have explored the different attributes in classifying learning styles. In classification process many algorithms have been explored amongst which Bayesian network and Rules-Based has outperformed. The less focused areas revealed in this paper was learning styles through games and other media sources. Although the implementation of gaming in learning styles has shown better results it has not been further explored. (Truong, 2016)

MERN Stack Web Application

The application developed using MERN stack is a single page responsive application. The front-end of this application is built using ReactJS which is a library of NodeJS. The functionalities of this system is handled by NodeJS. It consists of methods and components that efficiently handles all the functionality of this system. The backend server is provided by ExpressJS, a library that acts as server for the software. Moreover, for storing data MongoDB is used which is responsible for receiving http request from the client that is send to the server and the server fetches the data and sends the required information to the user. This bond of communication between client and server in MERN is formed by axios. The fetching of data from the database is provided by Mongoose Object Documentation Map for MongoDB.

The application has sign up form that asks the user to enter their username, password, email, phone number which gets validated and stored in the database after the password is hashed. After the successful registration the user the user is sent a notification via email. Then the user can log in to the system using their username and password which gets validated from the database. The system is based on team members assigning tasks to members and the communication between the task giver and the task holder is formed by this system. The task giver can assign task to the member by providing short description on the task with the deadline as well as they can edit, delete the task as per the necessity. Whereas, the user are allowed to view the work they are assigned change the status of completion and add their own work on the assigned portal as well as change their password.

The motive to develop a well performing web application using MERN is to collaborate task management for a team. This system manages the work assignment properly to each members with proper functionality that makes the communication between the work creator and members easier. The main communication medium is email where every time the job is edited or assigned each and every one using the software is notified. It creates an easy and effective communication between the team.

ReactJS utilizes JavaScript language maintained with the help of Facebook. MongoDB is a schema-less database that produces results in JSON format. MERN stack is overall based on JavaScript language which is a widely used programming language and comes handy when we

compete for job market. Express JS acts as the backend server that outlines the routes and arranges the HTTP requests according to the users' specification. (Raju, et al., 2021)

Sololearn

Sololearn is an e-learning website that provides free learning environment for the information technology enthusiasts. It provides varieties of courses for its students so that they can learn coding effectively and efficiently.

Firstly, the users will get an insights of the benefits on learning how to code and the system asks them to register if they do not have an account or log in if they already have an account.

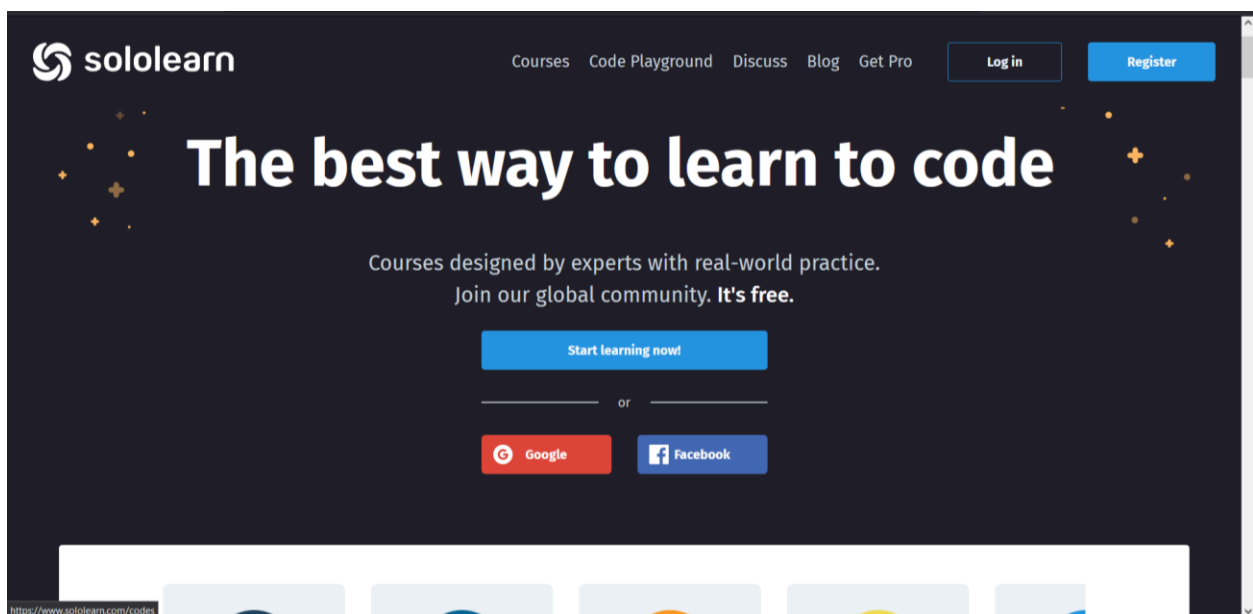


Figure 3: UI of Sololearn Homepage

If the courses are accessed without signing in the system automatically redirects the user to the sign in form and ask them to sign in before going further with learning.

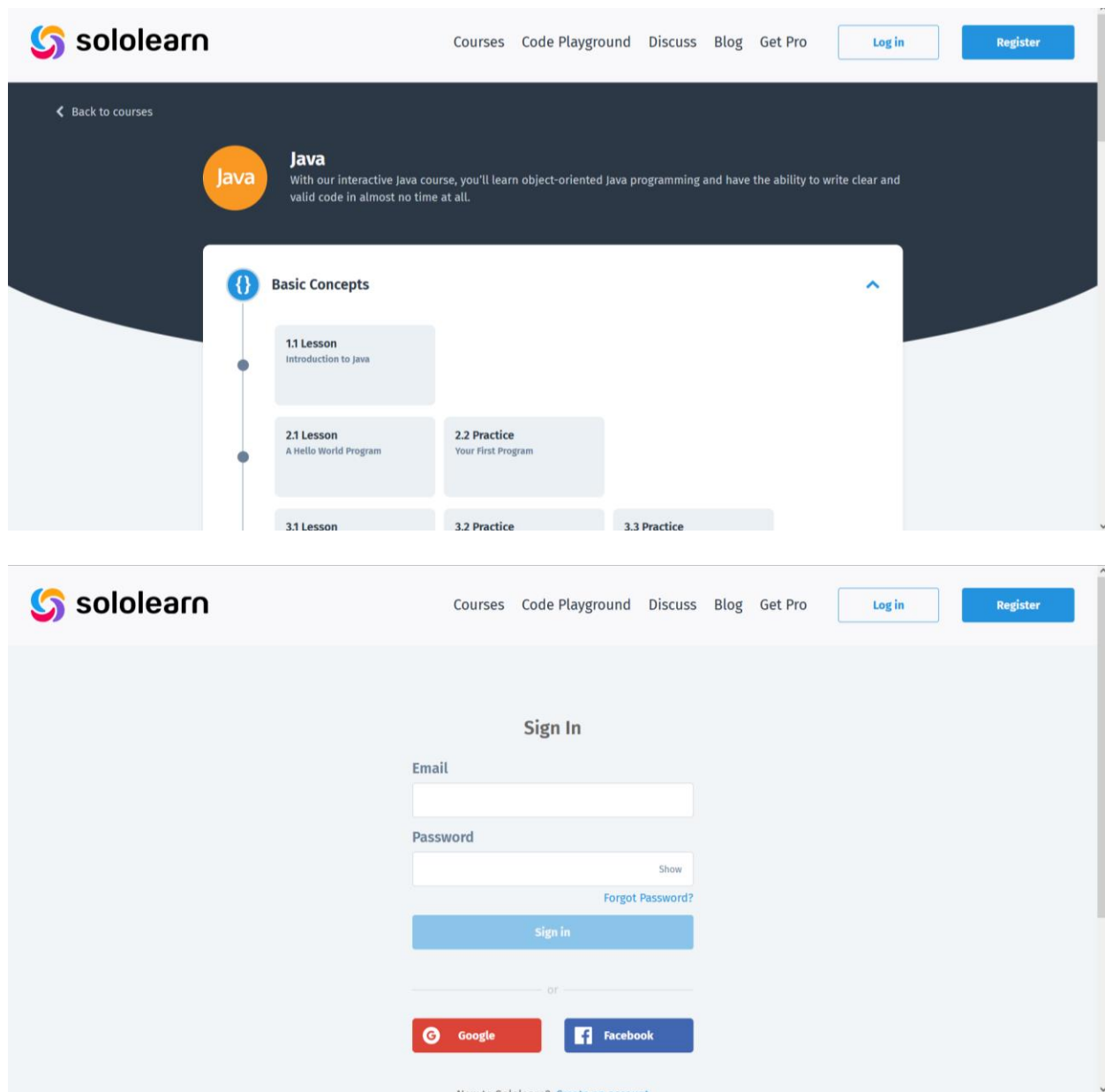


Figure 4: Sign in form of Sololearn

Sololearn allows the user to register through their google account or Facebook account. After registering it successfully makes an account of the users. The users are allowed to add their picture on their profile. The users' profile comprises of the courses the user is currently studying or in progress, their recent achievement, certificates that they have achieved on successfully completion of a course, challenges they can participate and the codes they have created while learning a course.



Jeshmi Rajak

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Here is where you left off

Lesson

The for Loop

Continue learning

Courses Progress

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Courses Progress

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C#

20%



Data Science

15%



HTML

68%



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Recent Achievements



Next task



Course Master

Complete 10 courses

[See More](#)

Certificates

[<](#) [>](#)



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JavaScript for Beginners

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Hey! You missed on some challenges.



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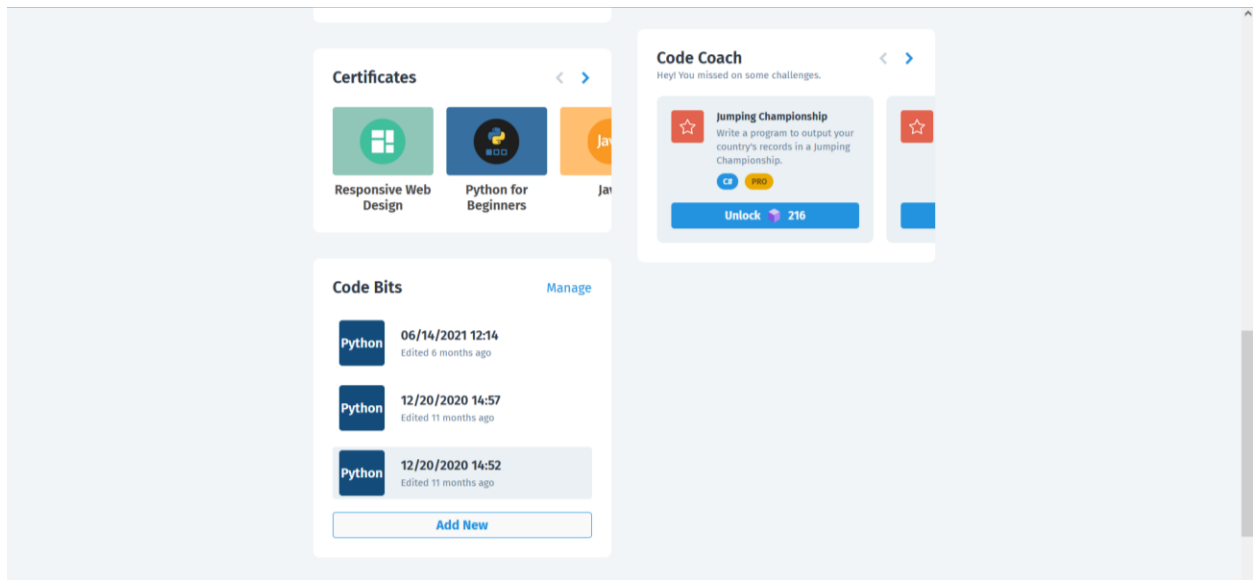
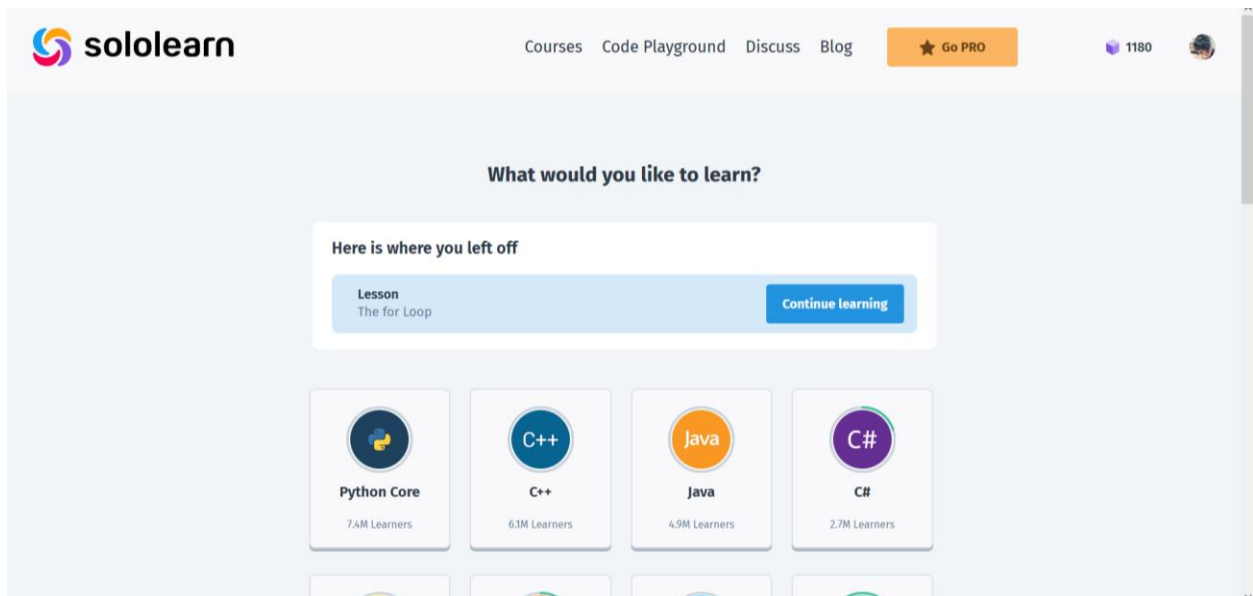


Figure 5: User Profile in Sololearn

The courses available in Sololearn are all coding based which are very helpful for students pursuing career in information technology. It develops self-learning abilities in students and make them competitive in the job market.



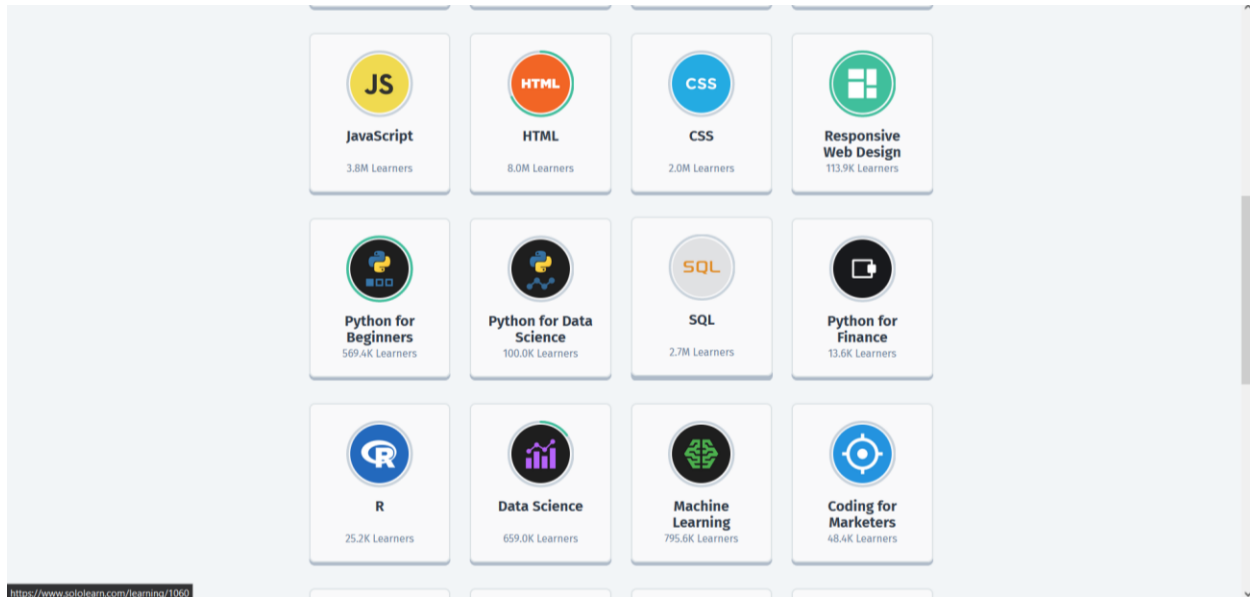


Figure 6: Courses in Sololearn

The courses provided by Sololearn makes the students learn from basic concepts to core concepts of each programming language. It helps the students to learn and cover each and every topic in detail that they should know before starting to code.

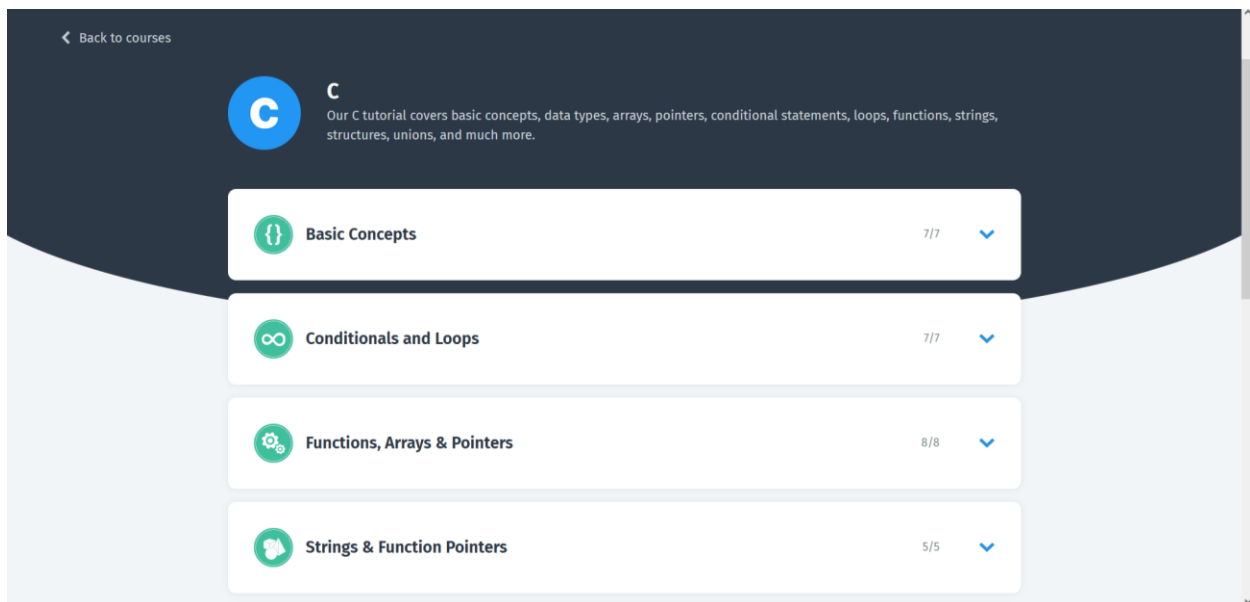


Figure 7: Course offerings

Each course is further divided into sub topics where each lessons are covered in detail with important notes highlighted below in case the information gets missed out.

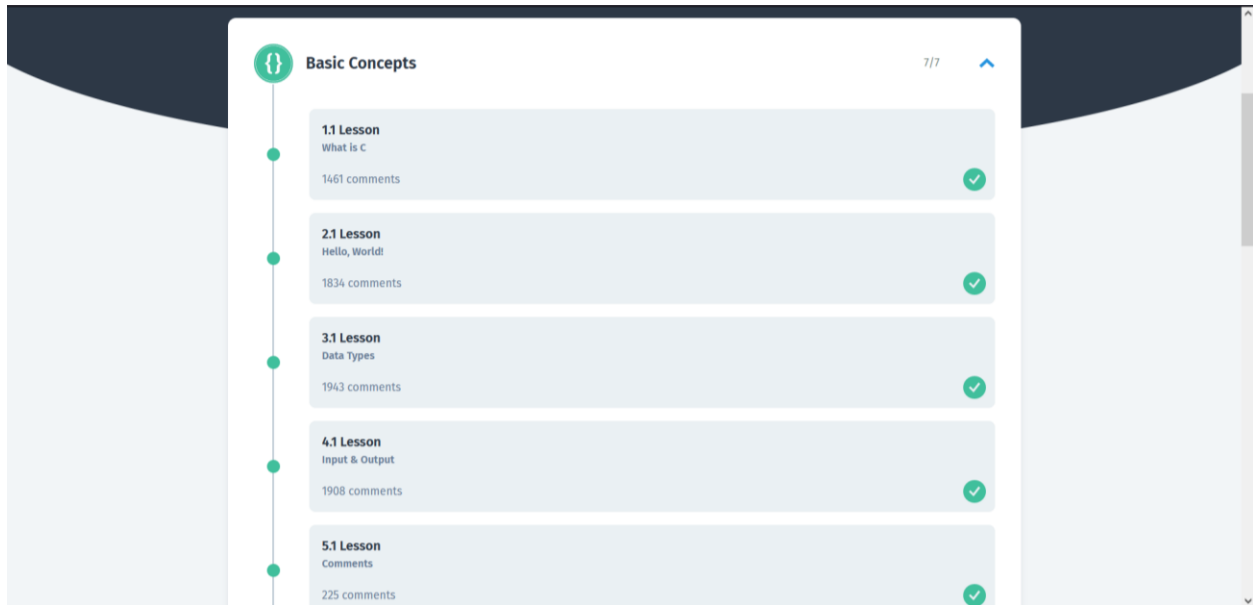


Figure 8: Lessons of each course in Sololearn

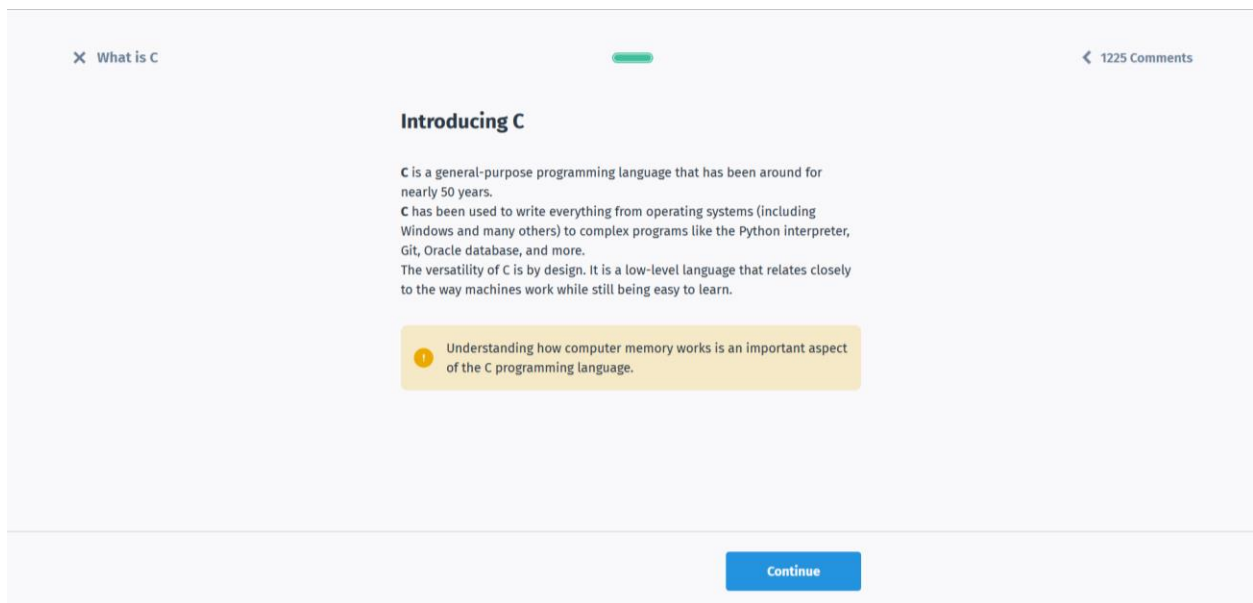


Figure 9: In-depth lessons for each topic

After learning the lesson in detail we can continue further to access other lesson where we are redirected to a quiz user interface where we need to answer the correctly to keep moving forward. This makes the learning effective as right after gaining some knowledge the students gets to revise it through playing quiz games. On clicking the answer and check button it automatically reviews

the answer and gives green signal if correct and red if incorrect. After getting the answer correct points are earned and the user is directed to continue learning next lesson. Here, the lessons gets unlocked as per the users completes the lesson successfully step by step.

✕ What is C

236 Comments

Introducing C

C is a:

Photo editing program

General purpose programming language

Client-side scripting language

Back

🔒

Check

✕ What is C

236 Comments

Introducing C

C is a:

Photo editing program

General purpose programming language

Client-side scripting language

Back

🔒

Check

Introducing C

C is a:

Photo editing program

General purpose programming language

Client-side scripting language



Well done!

Back



Continue

Introducing C

C is a:

General purpose programming language

Photo editing program

Client-side scripting language



Incorrect, try again!

Back



Retry

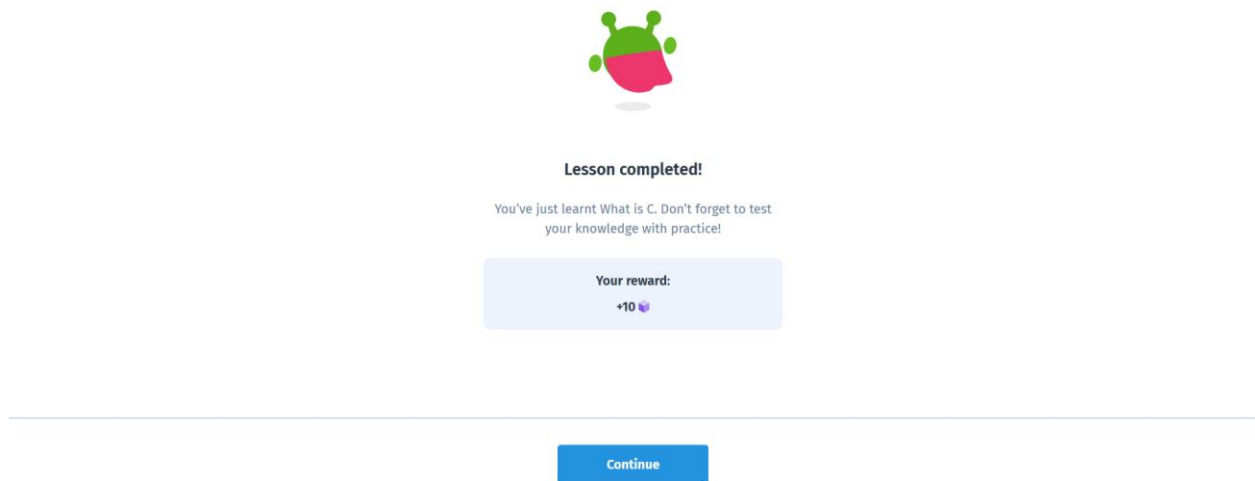
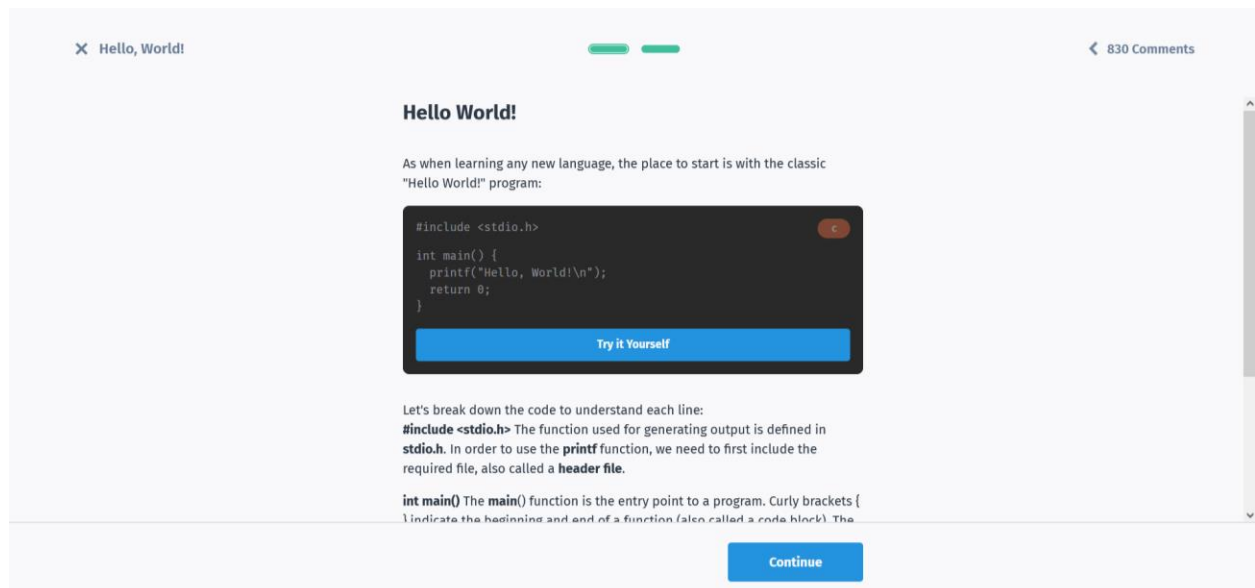


Figure 10: Quiz game in Sololearn

The coding lesson in Sololearn are taught in this manner when the try it yourself button is clicked the user is redirected to other page. This page comprises of compile time environment and runtime environment. The most amazing thing about this page is that it executes various codes according to the course and there is no need for a text editor to actually learn coding.



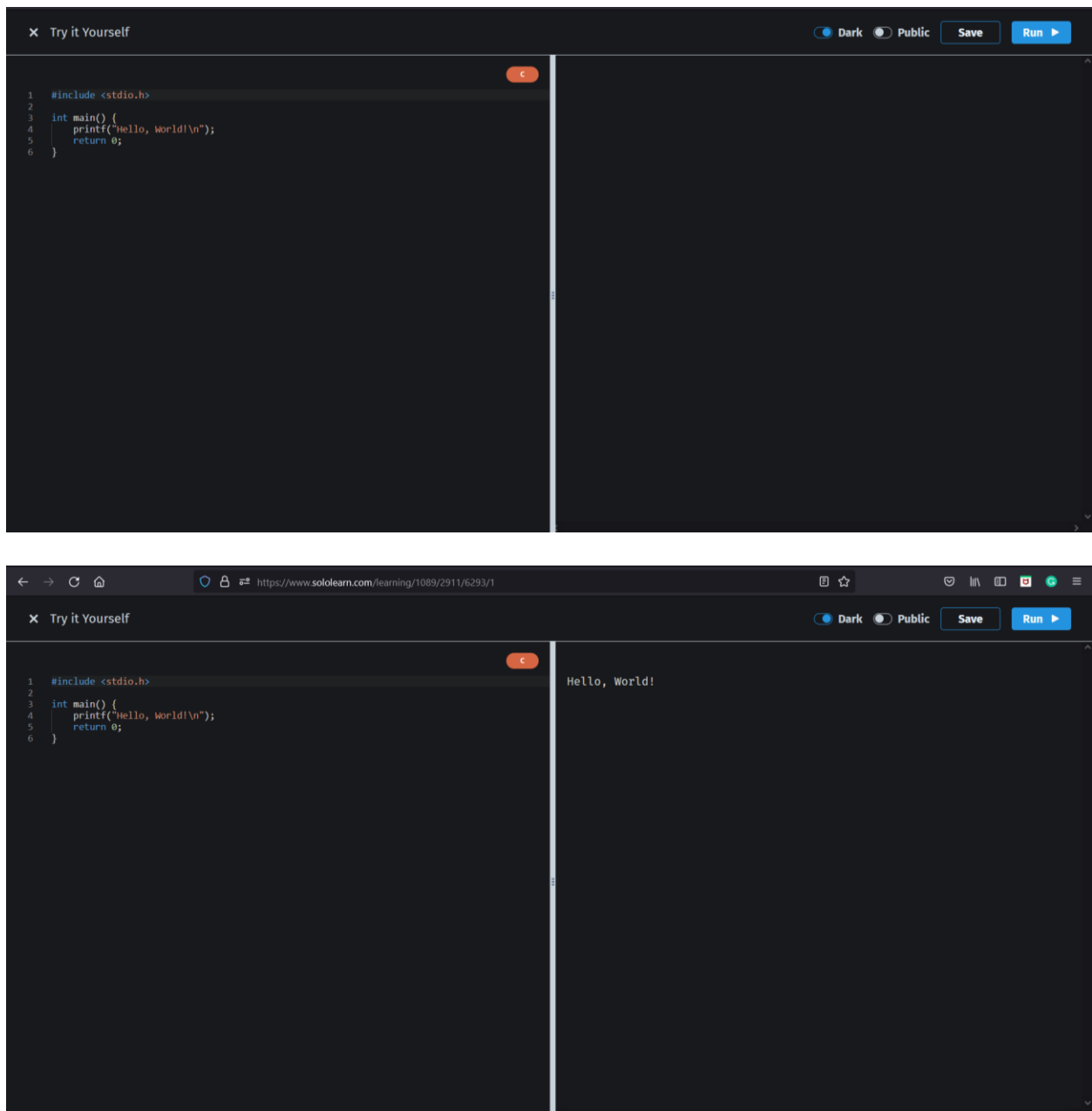


Figure 11: Text editor in Sololearn

On successfully completing the course, the certificate provided in the bottom gets unlocked and the users can claim their certificate for that course.

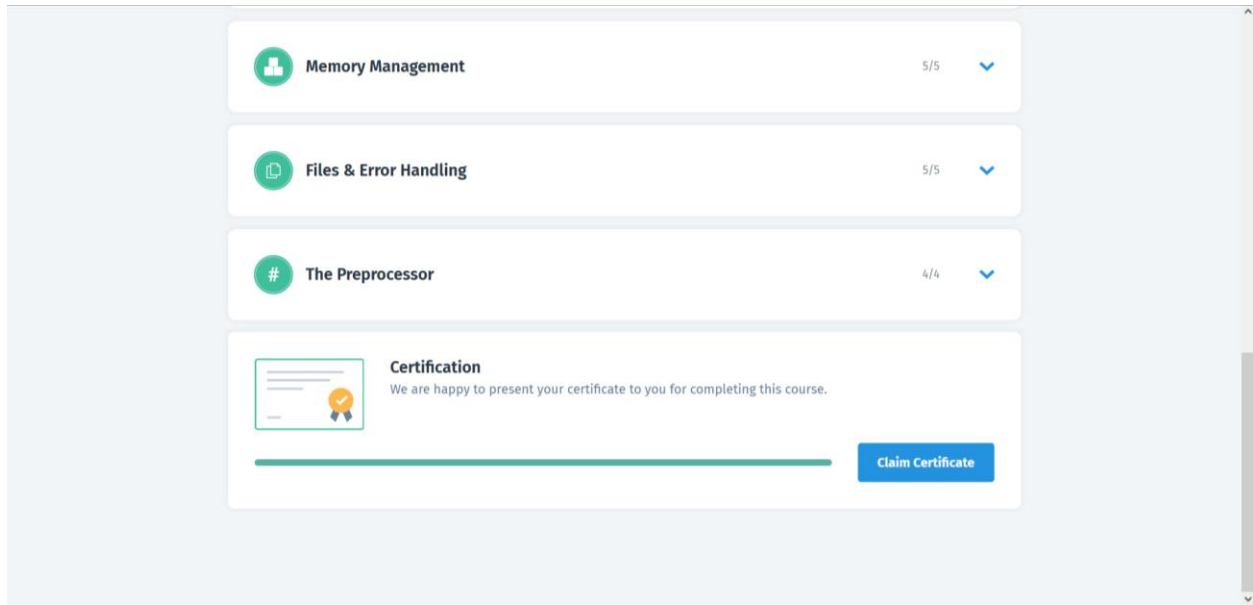


Figure 12: Display of certificate after completion of course

On claiming the certificate a pop out page appears where an interface with users' name and what course they completed is written and claims that the user is certified and qualified in that programming language.

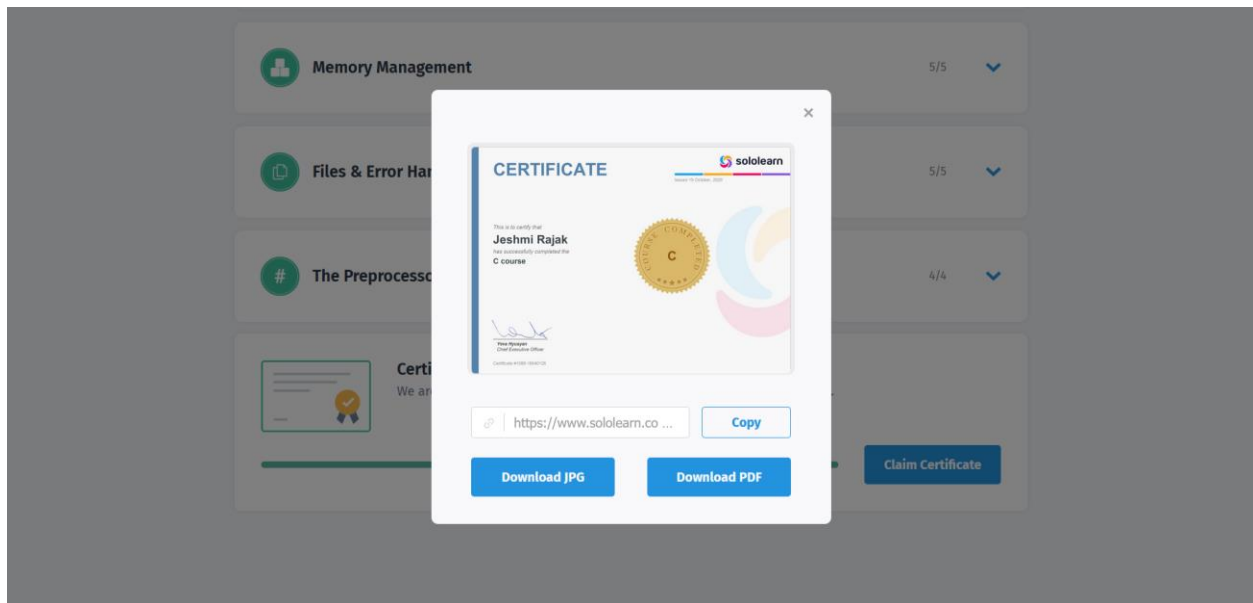


Figure 13: Certificate claimed in Sololearn

The page called the code playground allows the users to share their codes and compete with other users in the world of coding. It helps to share knowledge and gain experience from one another building communication skills that comes handy in work environment later on.

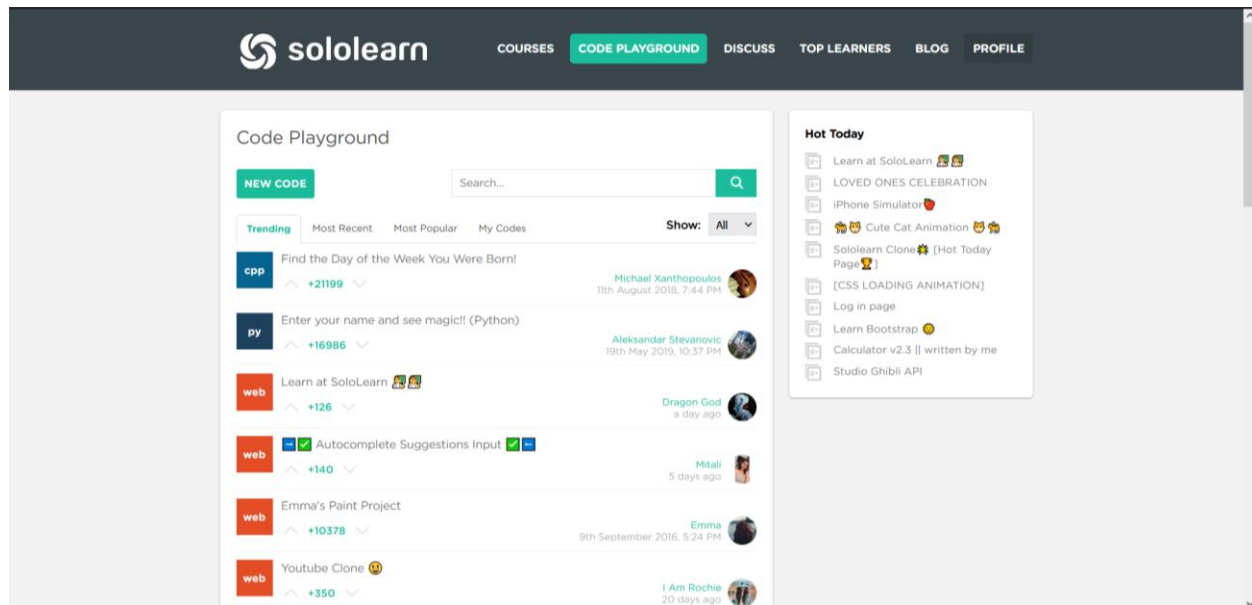


Figure 14: Code sharing playground

The discuss page in Sololearn is like a Q&A page where many learners share their confusion and get relevant answers from other learners. This way it makes the learning smooth and easy for the students.

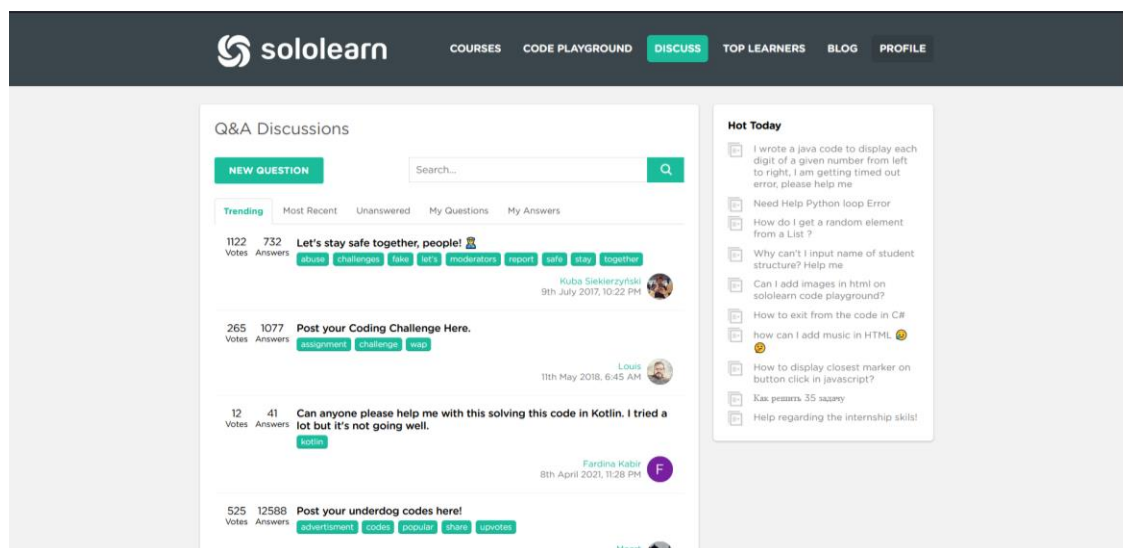


Figure 15: Discussion page in Sololearn



sololearn

[COURSES](#)
[CODE PLAYGROUND](#)
[DISCUSS](#)
[TOP LEARNERS](#)
[BLOG](#)
[PROFILE](#)

By Course

























Global
Leaderboard

1



JOY

1930992 XP

2



JEMΔ 🇧🇷 🇵🇹 [ActiveChallenger]

1545506 XP

3



MiKoLa 🇮🇹

1306472 XP

Blog section includes the information on topics that are very renowned and as a student of computer science needs to know about it. It provides details on each topics and gives insights on this topic with opportunities for the learners in this field.

28

DECEMBER 03, 2021

Cybersecurity Careers For Coders

Cybersecurity is a hot-button career topic these days. With more and more bad actors pulling off major cyber attacks and virtual crime sprees, the need for good guys who can step in and protect companies, critical infrastructure, and even national governments from hackers continues to grow.

Whether you're just starting your learn-to-code journey or you're looking to level up your coding career, cybersecurity is an excellent choice. In this article, we'll explore several cybersecurity career options that require some coding experience. You'll find out who's hiring for these jobs and how you can shift your coding knowledge into career-building cybersecurity skills.

Coders In Security Operations

When you think of cybersecurity careers, you may first think of ethical hacking -- being paid to test the defenses of networks and software against potential

Figure 17: Blogs in Sololearn

The use of this system really develops coding skills in students as well as provide them lots of opportunities in career development. The e-learning system is really helpful and reliable when it comes to pursuing skills online during free time.

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Figure 18: Career development recommendation in Sololearn

Khan Academy

It is an e-learning platform founded by Mr. Sal Khan. Khan academy provides free education for every age group with various courses to offerings such as: - mathematics, history, art history, science, computing, mock-test preparation for SAT, etc. It allows the students to learn at their own pace through various educational videos for topic they want to learn about as well as helps them to clarify their confusion with the help of coaches made available by this system.

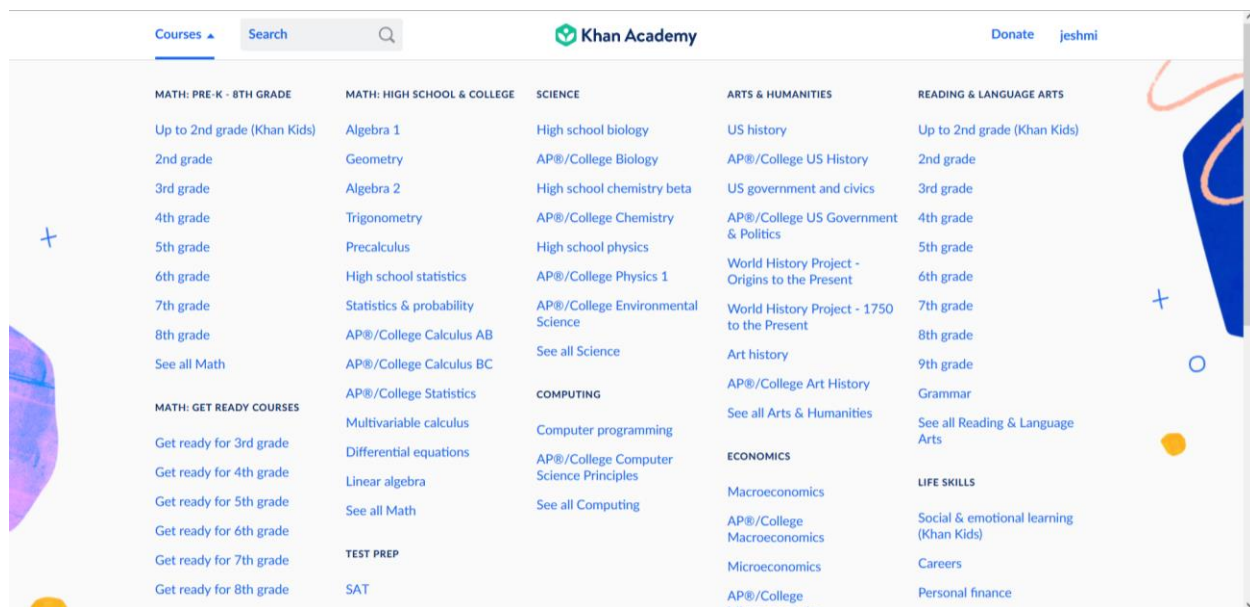


Figure 19: Course offerings in Khan Academy

Each users can create their account via mail, Facebook, apple id to access and use this web application. The courses a user takes are saved on their profile and they are notified accordingly each day to practice and continue what they are learning. Likewise, a user's profile gets updated with their progress in their pursued course and their achievement in that course are saved and displayed on their profile. Users can also join a teachers' classroom or select a coach to help them clear their queries.

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RESUME LEARNING

SAT
Test prep

Heart of algebra
SAT strategy archive: Math

Problem solving and data analysis
SAT strategy archive: Math

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Latest activity may take 10 mins to show below.

Last 7 days

All content

All activities

0 exercise min

1 total learning...

ACTIVITY	DATE	LEVEL	CHANGE	CORRECT/TOTAL PROBLEMS	TIME (MIN)
Solving linear equations and linear ... SAT	Dec 12, 2021 at 7:52 PM	-	-	-	0
Math SAT	Dec 12, 2021 at 7:49 PM	-	-	-	0

31

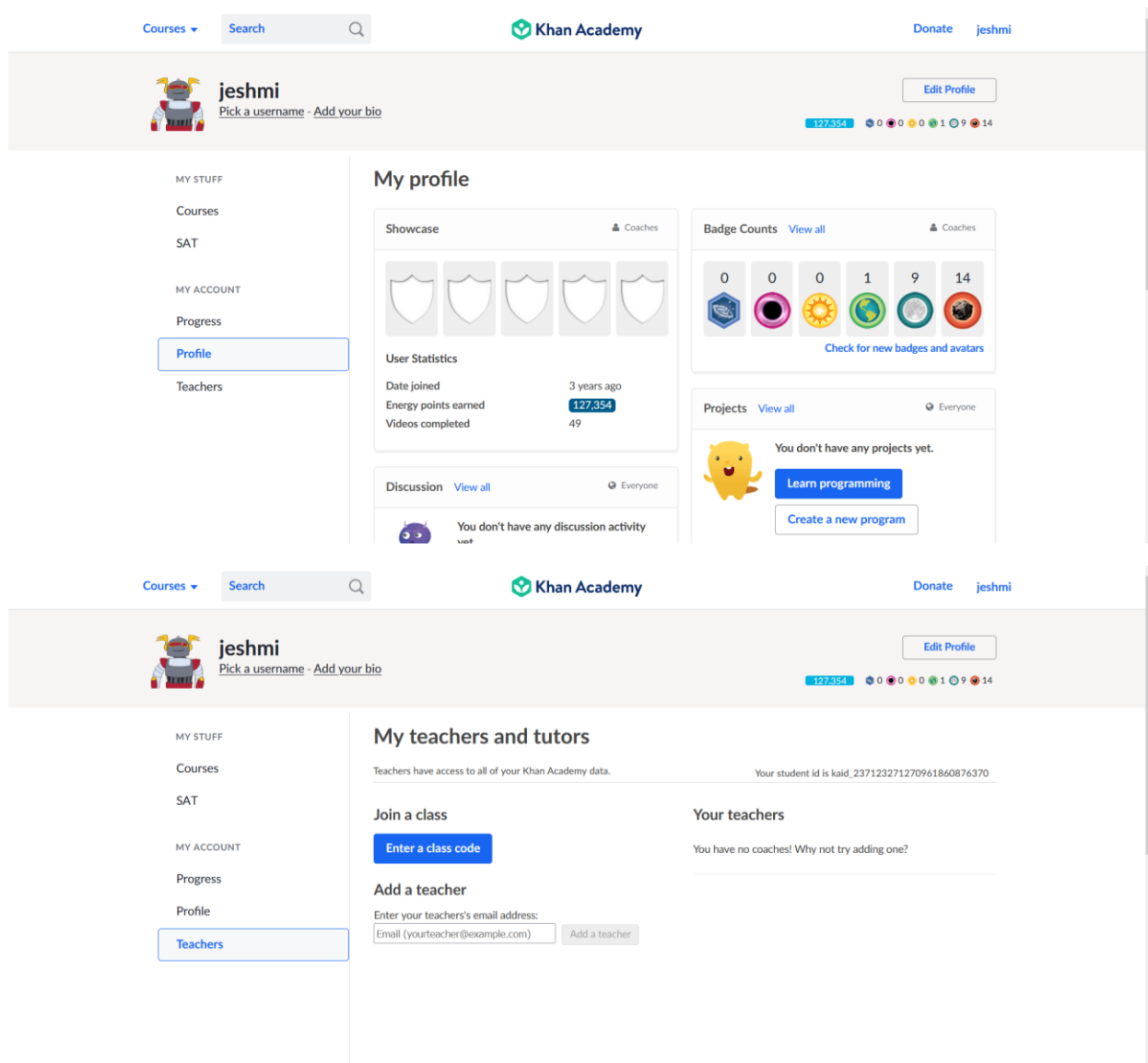


Figure 20: User profile in Khan Academy

The following dashboard is of SAT course which is personalized for every students. The mock test results for each test the student has taken is provided in detail which helps them to mark themselves in which area they need more practice.

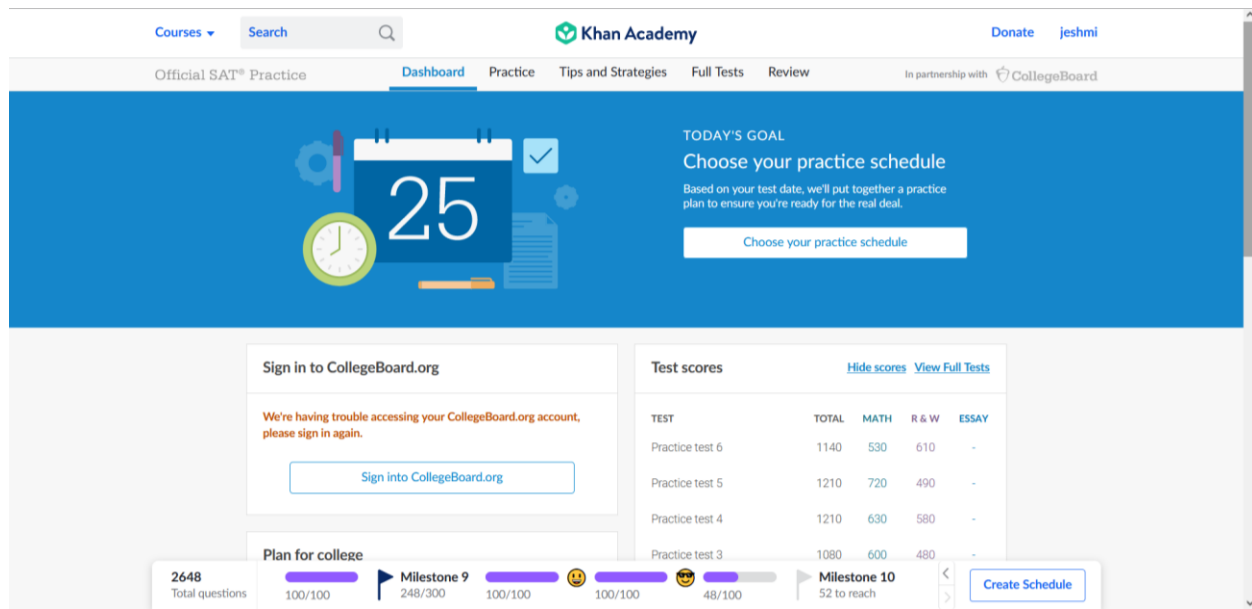


Figure 21: Personalized course dashboard of student

This application even records the weak point of the learners from previous practice and suggests them to work more on that particular topic so that they can sharp their skills properly and learn efficiently without any hurdle. The recommendation of this system is really well developed as a student it is very engaging and helpful.

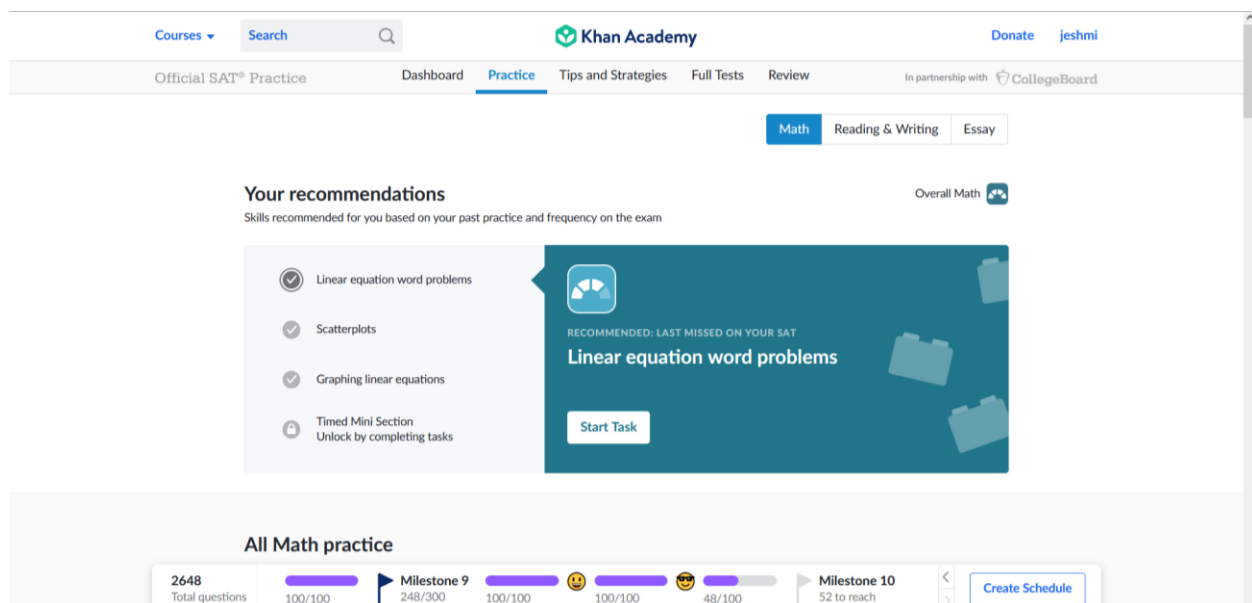


Figure 22: Topic recommendation

The following are the lessons that SAT course offers. Likewise, other courses also provides lessons and practice question in this manner.

All Math practice

Heart of algebra
These skills focus on linear equations, their graphs, and their applications.

How-to examples

Topic	Practice	Basic example	Harder example
Solving linear equations and linear inequalities	Practice	Basic example	Harder example
Interpreting linear functions	Practice	Basic example	Harder example
Linear equation word problems	Practice	Basic example	Harder example
Linear inequality word problems	Practice	Basic example	Harder example
Graphing linear equations	Practice	Basic example	Harder example
Linear function word problems	Practice	Basic example	Harder example
Systems of linear inequalities word problems	Practice	Basic example	Harder example
Solving systems of linear equations	Practice	Basic example	Harder example
Systems of linear equations word problems	Practice	Basic example	Harder example

2648 Total questions

Milestone 9: 248/300 (100/100) 🟢

Milestone 10: 52 to reach (48/100) 🟡

[Create Schedule](#)

Figure 23: Lesson offerings in Khan Academy

On accessing the lesson, five questions are provided in multiple choice where a reader should study the question properly and choose the right answer. If the students have trouble solving the question then they have a choice to go through this chapter with the help of a video or just get a hint or report the problem and get their queries resolved. This helps a lot for students to understand the topic as well as get a core concept on how the problem is solved visually as well as theoretically. It creates an environment for students with proper classroom like discussion with a coach explaining the chapter on the background virtually.

QUESTION 1 OF 5

Video Hint Report

$$y = (x - 2)(x + 8)$$

The equation above represents a parabola in the xy -plane. Which of the following equivalent forms of the equation displays the y -intercept of the parabola as a constant or coefficient?

☐ A $y = x^2 + 6x - 16$...

☐ B $y = (x + 3)^2 - 25$...

☐ C $y + 24 = (x + 2)(x + 4)$...

☐ D $y + 25 = (x + 3)^2$...

Do 5 questions: 1 2 3 4 5

Check answer

Figure 24: Multiple choice answers on Khan Academy

The multiple choice answer allows a student to choose and review their answer. The review shows correct and green when right and incorrect and red when wrong. The learners gets some hint below the answer if they get it incorrect and can make corrections accordingly.

QUESTION 1 OF 5: CORRECT

Video Explanation Report

$$y = (x - 2)(x + 8)$$

The equation above represents a parabola in the xy -plane. Which of the following equivalent forms of the equation displays the y -intercept of the parabola as a constant or coefficient?

☒ A $y = x^2 + 6x - 16$

For $x = 0$, $y = -16$.

This choice is an equivalent quadratic equation in standard form, which displays the y -intercept of the parabola, -16 .

☐ B $y = (x + 3)^2 - 25$

For $x = 0$, $y = -16$.

This choice is an equivalent quadratic equation in vertex form. It does not display the y -intercept, -16 .

☐ C $y + 24 = (x + 2)(x + 4)$

For $x = 0$, $y = -16$.

This choice does not display the y -intercept, -16 .

Do 5 questions: ☒ 1 2 3 4 5

Next to question 2

Manipulating quadratic and expo... Khan Academy Exit this Practice

QUESTION 2 OF 5: **INCORRECT** Video Explanation Report

$f(x) = x^2 - 4x - 1$

Which of the following is an equivalent form of the function f above in which the minimum value of f appears as a constant or coefficient?

A $f(x) = -(x-2)^2 - 5$
This choice is not equivalent to $f(x) = x^2 - 4x - 1$.
 $-(x-2)^2 - 5 = -x^2 + 4x - 9$

B $f(x) = (x-4)(x-1)$
This choice is not equivalent to $f(x) = x^2 - 4x - 1$.
 $(x-4)(x-1) = x^2 - 5x + 4$

C $f(x) = (x-2)^2 - 5$
The graph of f is a parabola that opens upward. The minimum value of f , therefore, is the y -coordinate of the vertex of the parabola.
This choice is an equivalent function in vertex form, which shows the y -coordinate of the vertex, -5 .

Do 5 questions: ☒ ☐ ☐ ☐ ☐ Next to question 3

Figure 25: Review answer on Khan Academy

It even provides full explanation videos for each chapter which helps the reader understand the topic in a more better way and energy points get counted the long the learners view the video that gets saved in the user's profile.

Manipulating quadratic and expo... Khan Academy Exit this Practice

MANIPULATING QUADRATIC AND EXPONENTIAL EXPRESSIONS – BASIC EXAMPLE

Watch Sal work through a basic Manipulating quadratic and exponential expressions problem.

Which of the following is an equivalent form of the function shown in the xy -plane, from which the x -intercepts can be identified as constants in the equation?

32 energy points

Choose 1 answer:

A $y = x(x+1) - 6$

B $y = \left(x + \frac{1}{2}\right)^2 - \frac{25}{4}$

C $y = (x-3)(x+2)$

D $y = (x-2)(x+3)$

Basic example Harder example

This choice does not display the y -intercept, -16 .

Do 5 questions: ☒ ☐ ☐ ☐ ☐ Next to question 2

Figure 26: Chapter Videos on Khan Academy

7. Project Methodology

The model that will be used is agile model. Agile model is preferred because the changes in the software development can be done in any point with continuous feedback from the client. This method includes both the iterative and incremental method. (Agrawal, 2019) Waterfall model is not preferred for the development phase as it is a linear approach and is feasible one single project that avoids scope changes once the project starts. (Santos, 2021)The product backlog I will be providing will give a clear view on the admin and user panel functionality.

7.1 Scrum

Scrum falls under agile method for software development. It is preferred than any other methods like waterfall because it is a iterative method and copes better with change to develop a software following all the necessary amends that needs changing for a project in a short time. It helps to divide what work to be complete in an allocated time by forming sprints to divide work until the project deployment phase. Since we are to be develop a project in a given time frame with a supervisor assigned to guide us through it scrum framework matched the software development methodology. According to scrum, development phase is done iteratively with flexible to changes needed and required to make that project reliable to the business we are targeting and supervisors are the ones making sure that out project is completed and market worthy during its deployment. (cprime, 2022)

7.2 Product Backlog

The product backlog is how scrum framework makes us categorize the features for our project. The following detail spreadsheet is shows the user story of how and what a user have to do to access the system and after doing so what they should be able to do in the system with the features priority stated. It acknowledges a developer the client's needs and priority and put their inputs accordingly making the development phase more effective and faster.

Product Backlog				
Epic	User Story	Acceptance Criteria	Priority	
Admin	I want to be able to add admin login so that I can add, update, delete users, lessons, recipes, images, videos, etc.	Given that I have added admin login page and when I enter username and password I need to be logged in as admin	Should be	
Admin	I want to be able to add admin sign up and admin login so that they can use the system	Given that I have added admin sign up and log in page when sign account must be created and after clicking log in the admin must be logged in	Must be	
Admin	I want to be able to add lesson in the course page so that the users can learn about the Barista course	Given that I have added lessons on the course page user must be able to view and make changes on accessing the page	Must be	
Admin	I want to be able to add fun quiz game after lesson so that the user can have fun while learning the course	Given that I have added quiz on the course page user must be able to play and choose answers	Must be	
Admin	I want to be able to enable and disable certificate for the users so that the users can claim it after the course is completed	Given that I have added certificate for the users admin must be able to enable certificate to those who have completed the course and disable certificate for those who have not completed the course	Must be	
User	I want to be able to create a user panel for user so that the user can take course provided by Kaldiabale	Given that I have added user panel the user must be directed to sign up to create account and an account must be created after signing in	Must be	
User	I want to be able to login after my account is created so that I can enroll and take the course	Given that I have added login page the user must be able to login by entering username and password and access the course offerings after enrolling	Must be	
User	I want to be able to take quiz so that lesson can be more clear and know whether the answer I have entered is correct or incorrect	Given that I have added take quiz for the user, the user must be able to play quiz and know whether the entered answer is correct or incorrect	Must be	
User	I want to be able to claim certificate so that I can use that certificate to earn my learned skilled	Given that I have added claim certificate the user must be able to claim their certificate after they complete the course	Must be	
Admin	I want to be able to add, delete video so that the learners can be more familiar about the techniques	Given that I have added the video the admin must be able to add and delete the video from the system	Must be	
User	I want to be able to play the video on the system	Given that I have access to the course the user must be able to play the videos on the system	Must be	
Admin	I want to be able to add, delete tools to the system so that the learners can be familiar about the tools used for training.	Given that I have added tools the admin must be able to add and delete tools on the system	Could be	
User	I want to be able to purchase the tools provided on the system	Given that I have added tools the users must be able to purchase them	Could be	

Figure 27: Product backlog

7.3 Sprint Backlog

Sprint backlogs help to maximize the allocated works to be done in a given time frame so that there will not be any delay in project submission and all the criteria so that by the client can be met until the given deadline. The backlog helps the developer to make the system as per the story given and finish it staying within the time stated.

Sprint Backlog									
Epic	User Story					Acceptance criteria			Priority
Admin	I want to be able to login so that I can create new admin assign them email and password, view all admins, delete admins.					I can create new admins so that they can perform the necessary changes in the e-learning system.			Could be
User	I want to be able to sign up and create an account in the e-learning system.					I can enroll in the course and learn what the course has to offer.			Must be
User	I want to be able to login in the system after my account is created in the system.					I can use the system just by entering email and password in case I get logged out of the system.			Must be
Admin	I want to be able to access the admin dashboard just by logging in.					I can use the system just by entering email and password in case I get logged out from the dashboard.			Must be
Admin	I want to be able to add lesson related to the course on clicking the lesson section of the admin dashboard.					I can create a lesson and that lesson is added successfully to the system where the learners can view and learn.			Should be
Admin	I want to be able to upload image and description of the that image.					I can upload images and description which must be updated in the system for the learners to view it.			Must be
Admin	I want to be able to add recipes for the new drinks.					I can create new recipe and add them on the system so that the users can learn and try making them.			Must be
Admin	I want to be able to update the quiz game in the system.					I can update the quiz questionnaires and answers so that the users can participate and revise what they have learnt.			Must be
Admin	I want to be able to add new quiz games in the system					I can create new games and the learners can play them to make their learning more efficient.			Could be
Admin	I want to be able to enable and disable certificate for the user.					I can enable certificate to the user when any progress of the user is seen and disable certificate in case no progress of the user is seen.			Must be
User	I want to be able to enroll the course on clicking the enroll button on the homepage after my account is successfully created.					I can access the course offerings with all the lesson that has to be mastered to be a good Barista.			Must be
User	I want to be able to participate in the quiz game.					I can participate in the quiz game so that I can revise the lessons I have learnt.			Must be
Admin	I want to be able to add tools for the system so that the user can get familiar about the equipments for the Barista learning.					I can add tools in the system to make the user familiar about them.			Should be
Admin	I want to be able to delete the tools.					I can delete the tools from the system successfully and the system gets updated.			Should be
User	I want to be able to purchase the tools shown in the system.					I can purchase the tools and the tools get added in my cart successfully.			Could be
User	I want to be able to claim the certificate after the completion of the course.					I can claim the certificate and view the certificate after claiming it.			Must be

Figure 28: Sprint backlog

7.4 Gantt chart

The tasks for the project started from research and according to the calendar ends with the presentation. Here, the start date and end date are determined based on the gap between two tasks as per the calendar. Likewise, day to complete is determined based on that.

Tasks	Start Date	Days to complete
Title Research	30-Aug	21
Title Submission	20-Sep	1
Project Proposal	17-Oct	31
Project Registration Form Submission	17-Oct	31
Literature Review	17-Nov	33
Code Submission	16-Nov	36
Development of the front end	16-Nov	36
Back end implementation	26-Nov	34
Artefact Design	20-Dec	29
Test-Plan	20-Dec	30
System Testing	16-Dec	17
Bug fixing	16-Dec	169
Professionalism Report	25-Jan	10
Final Report draft Submission	28-Feb	20
Final Report Submission	1-Mar	133
Presentation slides preparation	11-Apr	133
Poster Presentation	11-Apr	17

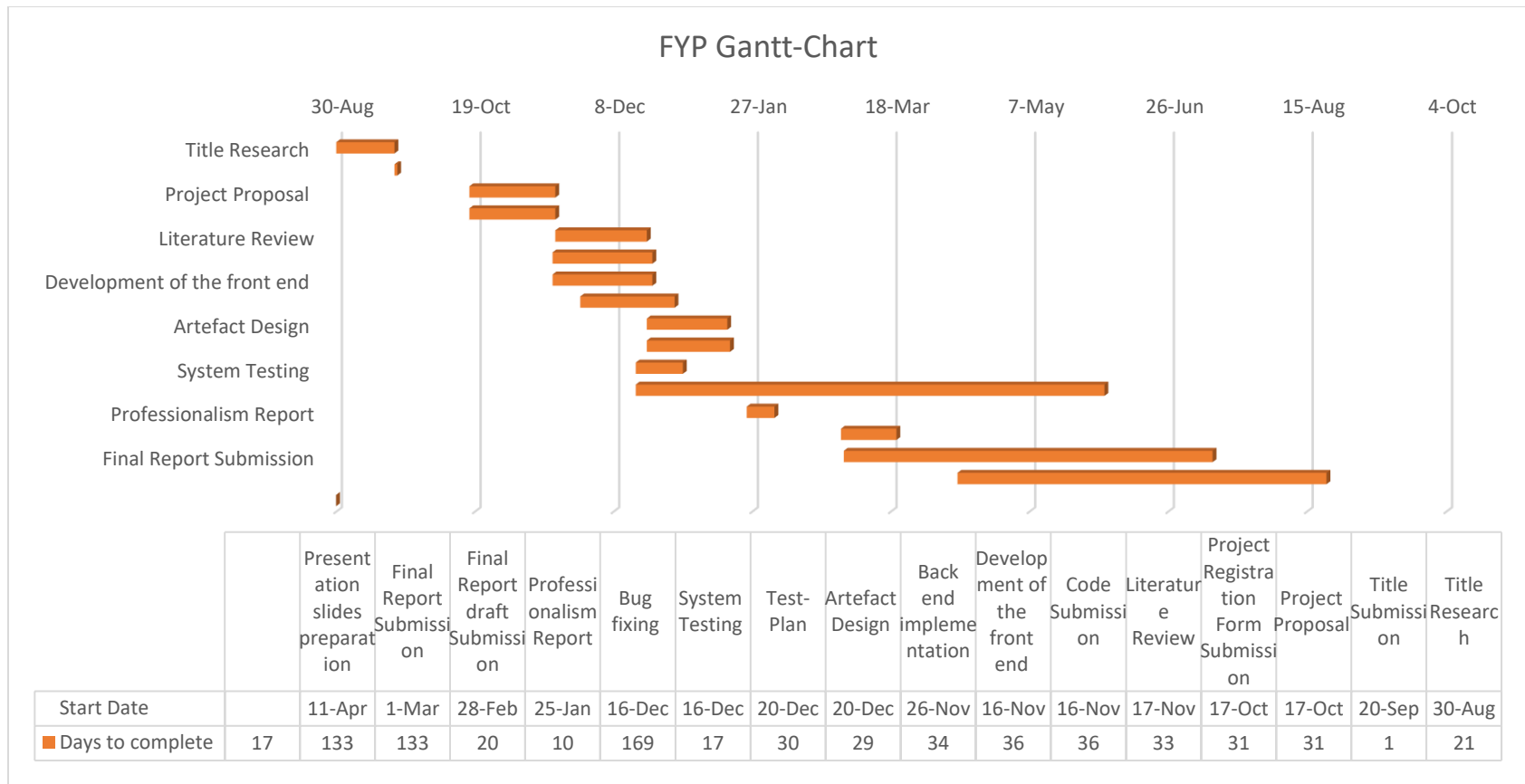


Figure 29: Gantt-Chart

8. Technology and Tools

Tools and technology used for the development of this project is briefly described below: -

- Agile model: The model is based on development of full software using iterative approach. The software development process is divided into smaller iterative steps that lasts for about 3-4 weeks. In agile method, the scopes and planning are decided before the start of the project. Therefore, use of agile approach reduces risks and time consumption during the development phase of the software. Agile method breaks down the development phase into five phases which are: - plan, design, develop, test and deploy. (javatpoint, 2021)
- Design: Figma software will be used for designing the wireframe. The color contrast for the website will be checked using A11y color contrast checker to increase the accessibility.
- Gantt chart: It is a graphical representation of tasks that needs to be done for completion of a project. This chart helps in scheduling and allocating time for each task therefore, it helps in understanding the dependency of each task with each other. It also helps the development process of a software well informed to the project manager. (The Economic Times, 2021)
- UML diagrams: It is Unified Modeling Language that helps to better understand the workflow of the system that an engineer is trying to design. The visual representation of the system provides standard way of modeling the flow of the work system as well as the features are also well categorized that helps the reader understand the concept of the proposed system. (Tallyfy, 2021)
- FDD: It is a function decomposition diagram for a business model or a system. It is a diagrammatical representation of how complex system function's work and the function interaction between various components. It helps during the project analysis. (tutorialspoint, 2021)
- ERD: Entity Relational Diagram is a representation of relationship with various entities or attributes. It is used mostly in the field of relational databases. It shows logical relation between each and every entities used in the database. (Lucidchart, 2021)

8.2 Programming Language

System is developed using MERN stack. MongoDB is used for storing data. Express.js is used for the server-side framework. React js is used for the front end and Node js is used for the backend. (MongoDB, 2022)

All these technologies are briefly described below: -

MongoDB - The database used for the storing purpose is MongoDB atlas. It is a schema-less database which stores data in the form of JSON format. It is different from other databases such as: SQL as it does not store data in the form of table. The advantage of MongoDB over other databases is it speeds up the application deployment with less complexity. The library used by MongoDB and Node.js is mongoose which is an Object Data Modeling library. It validates the schema provided for the system, as well helps to manage the data and translate the objects in code and helps displaying those objects for MongoDB. (Karnik, 2018) The following shows how data are stored in MongoDB database.



```
> {
  "_id": ObjectId("61e67848e78d0bd17e55ad91"),
  "fName": "Kristen",
  "lName": "Stewart",
  "password": "$2a$10$JLPu8.ch8P6L3S/ovyy1Nu1PgIm6cn/jnsC3DNaA2yhWURRjQWfuc",
  "confirmPassword": "$2a$10$0hpbNNuuIY3Rv4hRU9Un0G5RB/RUXFr.CadPE6fHc8v4mIPGGK96",
  "email": "kris@gmail.com",
  "phoneNumber": "345345454",
  "__v": 0
}

{
  "_id": ObjectId("61e68700995086bb28bdf82"),
  "fName": "Jeron",
  "lName": "Rajak",
  "password": "$2a$10$8z/0Csz..XajxJQVPA764uppi/te731zna/Pm5Ejv4n77sME6PCma",
  "confirmPassword": "$2a$10$0ORTHee4R1PG74xR9PLqG050xYYRbsMwtDyY0gFGRkJuJmKe2pqKC",
  "email": "jeron@gmail.com",
  "phoneNumber": "324234534545",
  "__v": 0
}
```

Figure 30: Data storing in MongoDB Atlas

Express.js – Express.js is used in this project to develop a web server using middleware that can handle all the request and responses that a user makes from the front end. Since React front end is a single page application express.js compliments it by specifying the routes used to preform

various functions dynamically based on defined URL. Also helps the users to redirect pages dynamically based on their performed action on that page. (Simplilearn, 2022)

React js - It is used to develop the user interface of the project. This technology provides a virtual DOM to render an app faster than any other DOM. It can be used on both the client as well as server side. (javaTpoint , 2021)

Node js – The server side in this project is handled by node js. Since it operates on single thread it can handle a hug number of requests and responses resulting in high scalability. Node js is different from any other web server techniques is they store each request in a new thread that results in taking up memory eventually running out of it. Whereas, node js stores it in a single thread where all the request made are concurrently stores in an event loop and the requests are met one after another without taking up extra space. (Denman, 2022)

8.3 Tools

Various tools used for this project are described below: -

IDE - Visual studio code is used as the text editor as it is compatible with almost all programming languages. It helps to correct and declare variables automatically while coding making it reliable to use. Moreover it is compatible with any platform supporting extensions for all languages without hampering its speed. Furthermore, it includes connection with Git for the timely and secure storage of the project. (Pedamkar, 2022)

Git – It allows to store the project as a backup online just by creating a repository. It helps to track our progress day to day by labelling commits that are meaningful that can help us mark changes as well as form a detailed chart of how long did we take for a project development. Additionally, it also allows us to add our supervisors as collaborator to track and evaluate our progress.

Postman - It is a tool that is used for API testing. It helps to organize API requests by collecting them in a folder with similar request sent with the built-in node.js runtime environment. It helps in every stage of API development. (Specbee, 2021) It helps to check whether the request made by the user is fetched or not also makes aware what kind of request is the user making such as: -

get, post, put, etc. It helps to encounter any error on the API. It helps to keep track the amount of requests made using the API.

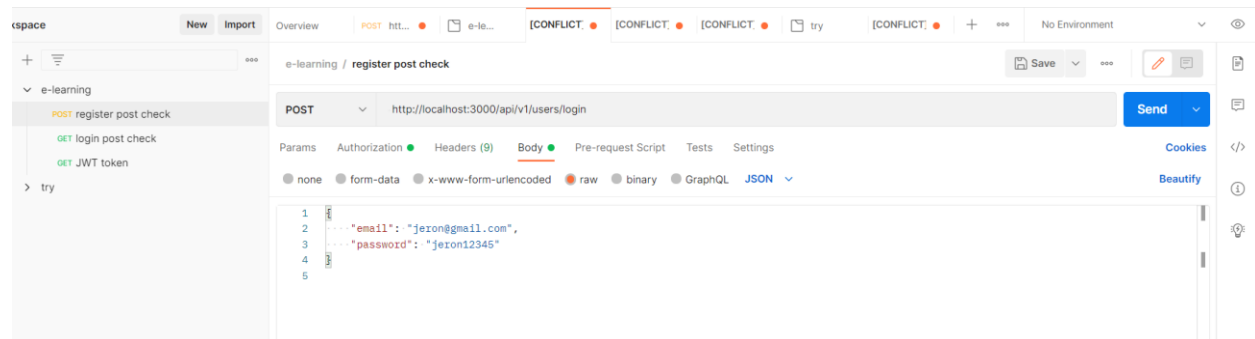


Figure 31: Testing APIs using Postman

8.4 Icon package and font family

Font awesome icons are used for the various icons in the project user interface like in the register, login page.

Font family used throughout the project user interface is Saira and Baloo 2 to maintain the consistency of the web application design.

8.5 NPM packages

Packages used in the client side are mentioned below: -

@material-ui/core, axios, clouinary, pdf-lib, moment, query-string, react, react-dom, react-helmet, react-file-base64, react-native-web, react-router, react-router-dom, redux, react-switch, redux-thunk, styled-components, react-scripts, react-icons.

Packages used in the server side are mentioned below: -

NPM packages- bcrypt, bcryptjs, body-parser, clouinary, concurrently, cookie-parser, cors, dotenv, express, express-fileupload, express-jwt, express-validator, jsonwebtoken, module, mongoose, Morgan, nodemailer, typemodule

9. Artefact Designs

Legend

Sub Systems

AMS :Admin Management System
UMS :User Management System
CMS :Course Management System
QMS :Quiz Management System
VMS :Video Management System
TMS :Tools Management System

Types of Requirements:

F :Functional Requirements
NF :Non-Functional Requirements
UR :Usability Requirements

Admin Management System (AMS)

Req. Code	Req. Description	Use case	Moscow Prioritization
AMS-F-1.0	The admin must be able to login to the system successfully and must be allowed to access the admin panel. [Access Admin panel]	Access admin panel	Must be
AMS-NF-1.1	Admin must be view the admin panel and view all the functionality they can perform using that panel.		
AMS-F-2.0	System should allow the main admin to create another admin to monitor and update day to day activities in the system. [create admin]	Create admin	could have

AMS-NF-2.1	The admin should be able to view the other admins.		
AMS-F-3.0	The main admin must be able to delete another admin. [delete admin]	Delete admin	could have
AMS-F-4.0	The admin should be able to check the progress of the users and based on their progress they should be able to enable and disable the certificate to the users. [enable or disable certificate to the user]	Enable or Certificate to the user	should have
AMS-NF-4.1	Admin must be able to view the progress of the users.		

Activity diagrams

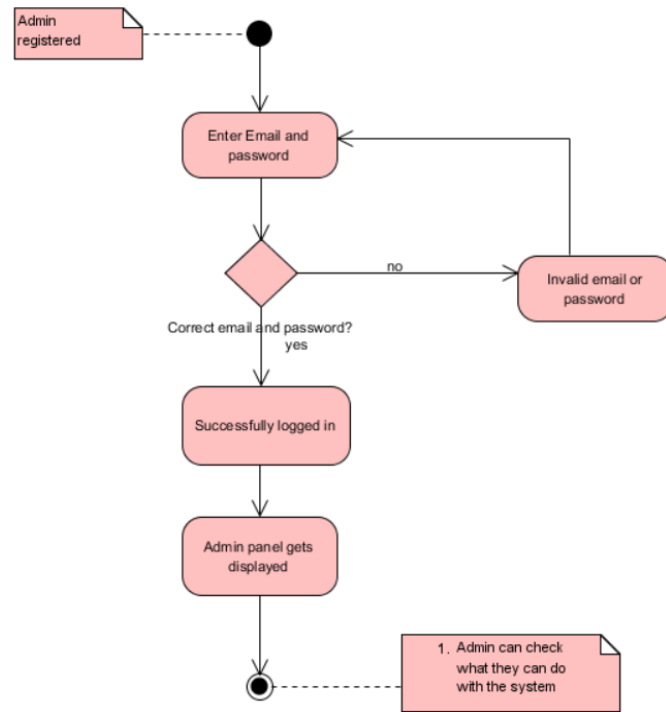


Figure 32: Activity diagram for admin login

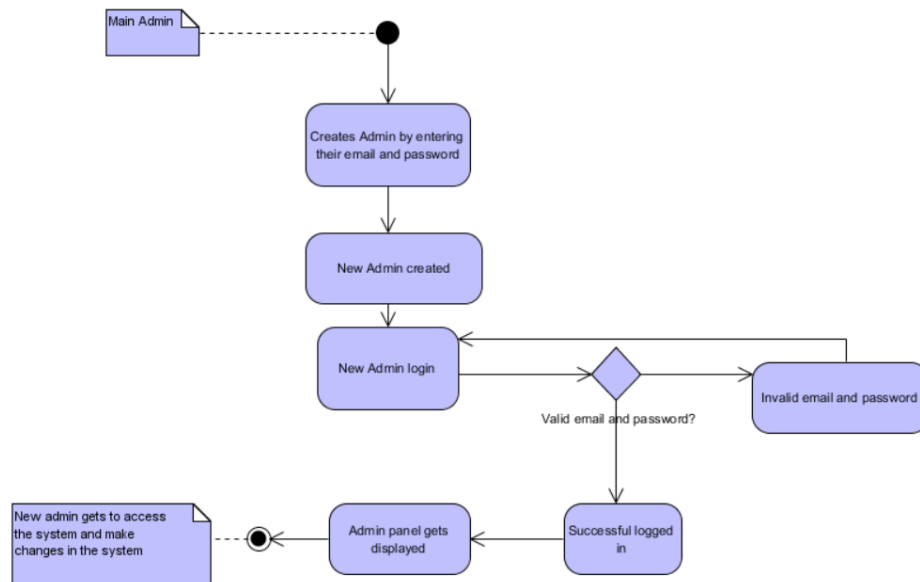


Figure 33: Activity diagram for adding new admin

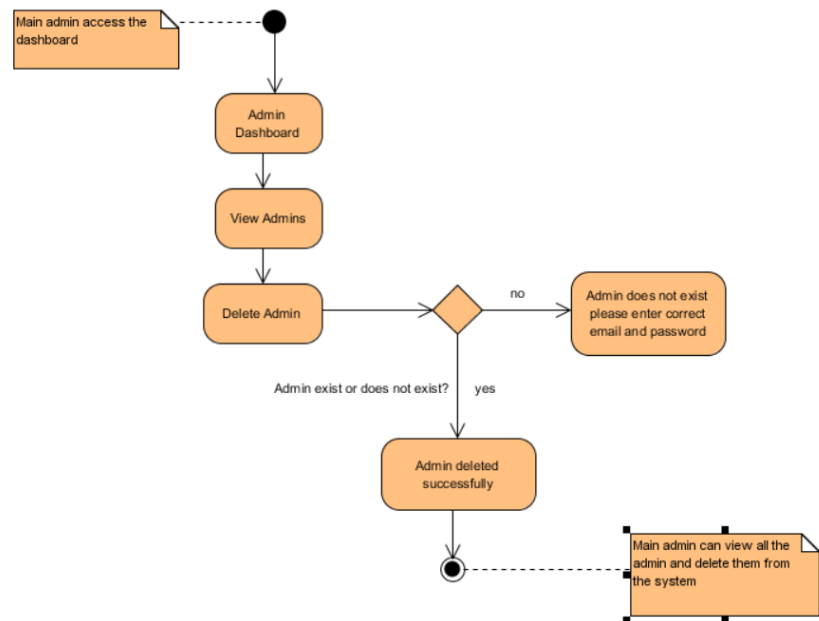


Figure 34: Activity diagram for deleting an admin

Use case Diagram



Figure 35: Use case diagram for admin management system

Sequence diagram

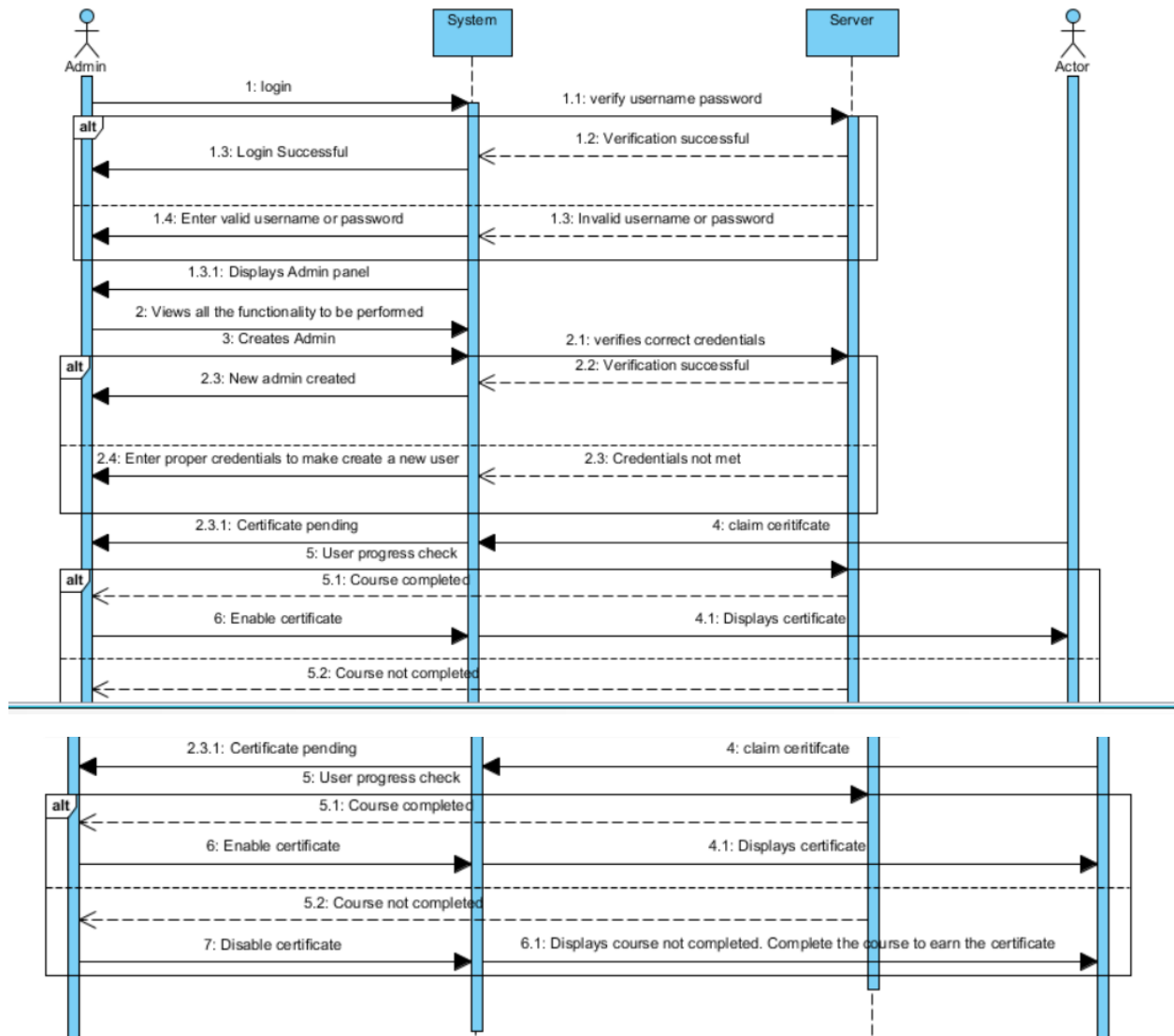


Figure 36: Sequence diagram for admin management system

Class diagram

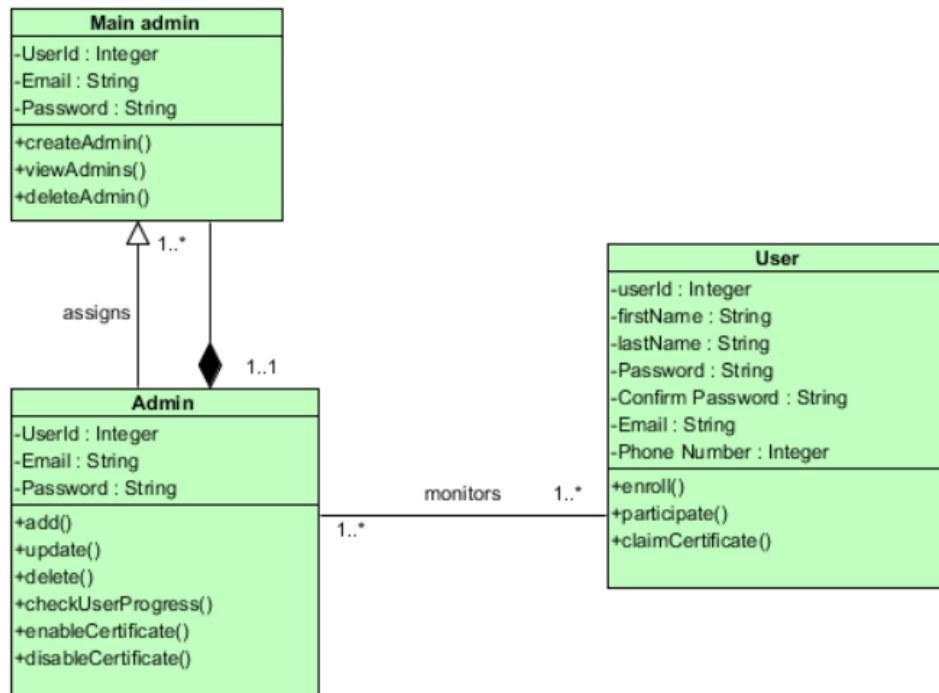


Figure 37: Class diagram for admin management system

User Management System

Req. Code	Req. Description	Use case	Moscow Prioritization
UMS-F-1.0	System should allow the user to sign up and create an account to use the e-learning system. [create user]	Create user	Must have
UMS-NF-1.1	While creating an account the user needs to make aware on which information should be filled where.		
UMS-NF-1.2	The users' password must be hashed after the user account gets saved in the server.		
UMS-NF-1.3	After signing up the user must be able to enroll and access the course.		
UMS-NF-1.4	After successful enrollment the user must be able to view the course offerings.		
UMS-UR-1.1	During the signing in process of the users leave a field not filled the system must alert the user to make sure they fill each and every filled in order to sign in.		
UMS-UR-1.2	The users must be able to access the system on any device.		
UMS-F-3.0	The users' progress on the course must be saved. [save progress]	Save progress	could have
UMS-NF-3.1	The users' must be able to view their progress on the course.		
UMS-F-4.0	The users' should be able to claim certificate after successful completion of the course. [claim certificate]	Claim certificate	Must have

UMS-NF-4.1	The user must be able to view the certificate.		
UMS-UR-4.1	The certificate should appear as a pop up to let the user know their certificate is claimed.		

Activity diagrams

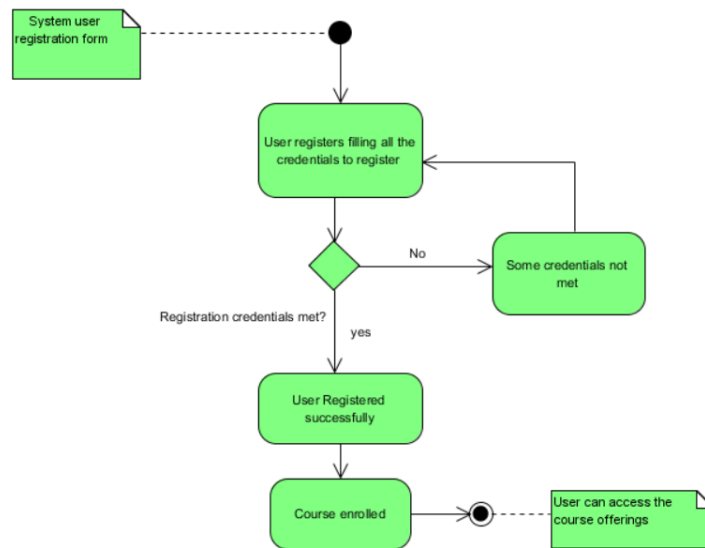


Figure 38: Activity diagram for user registration

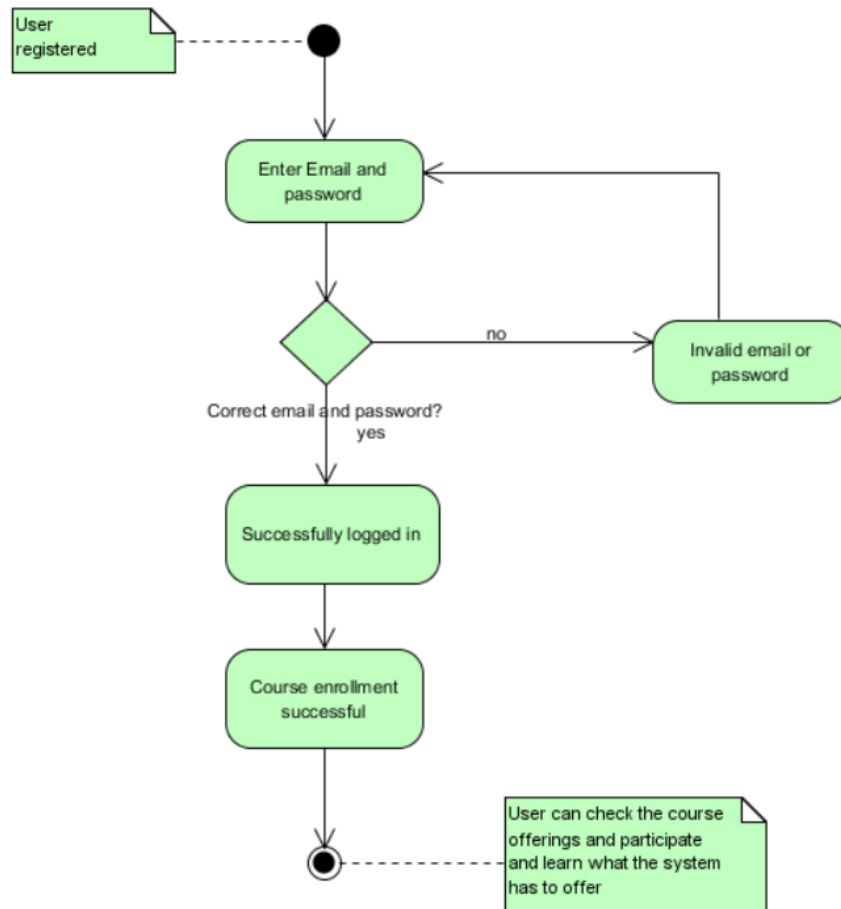


Figure 39: Activity diagram for user login

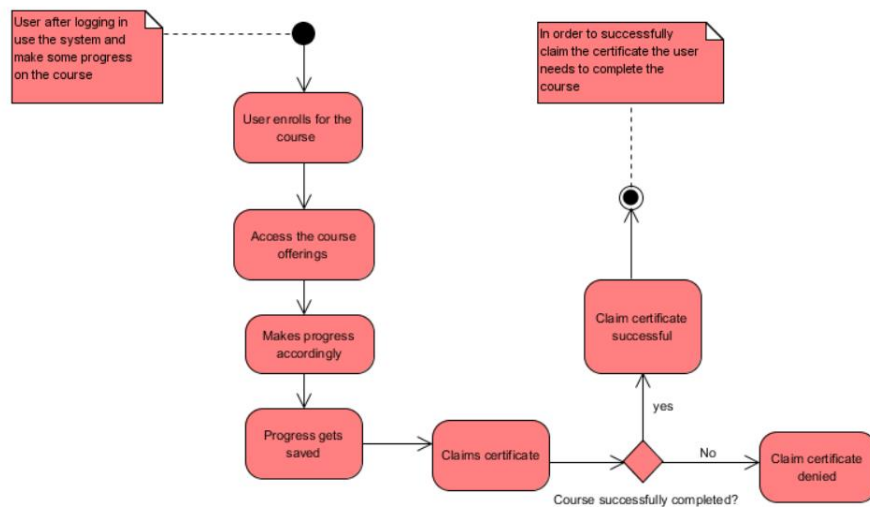


Figure 40: Activity diagram for user course enrollment activities

Use case diagram

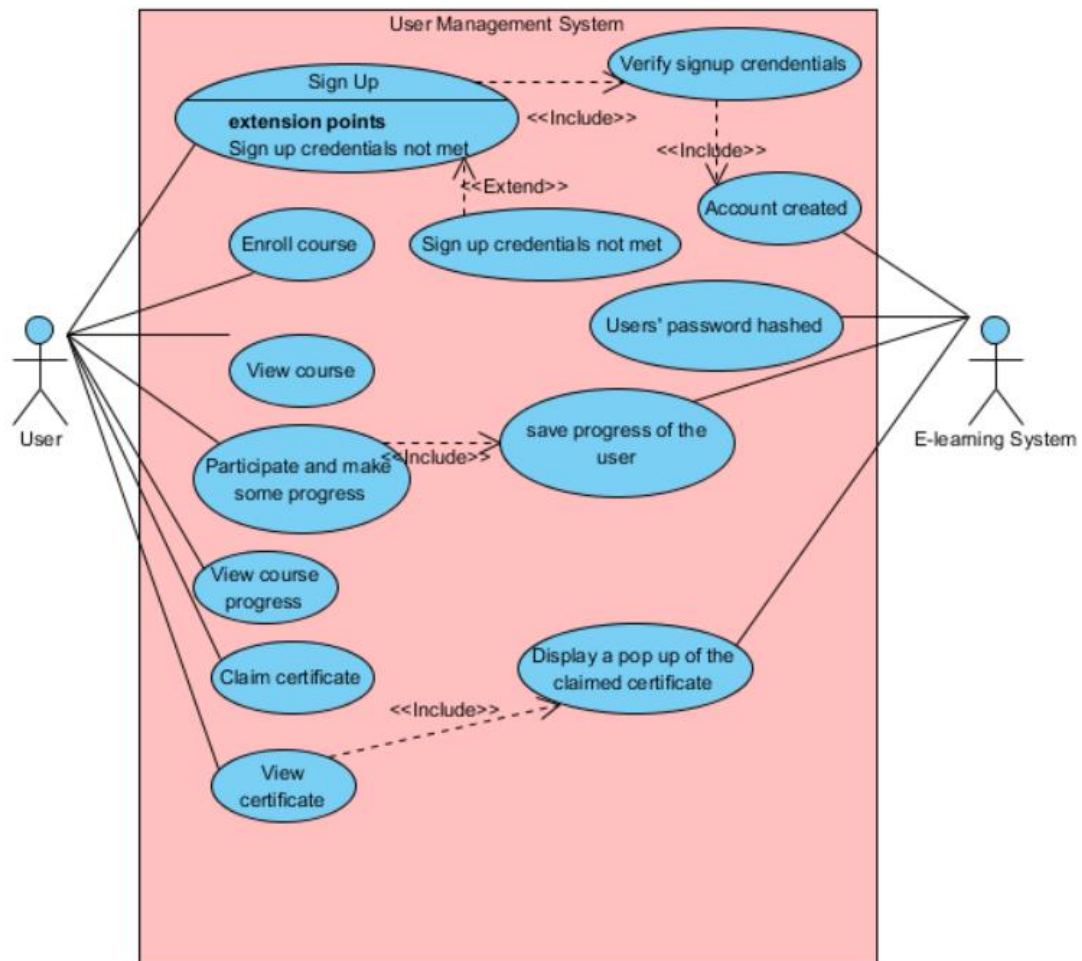


Figure 41: Use case diagram for user management system

Sequence diagram

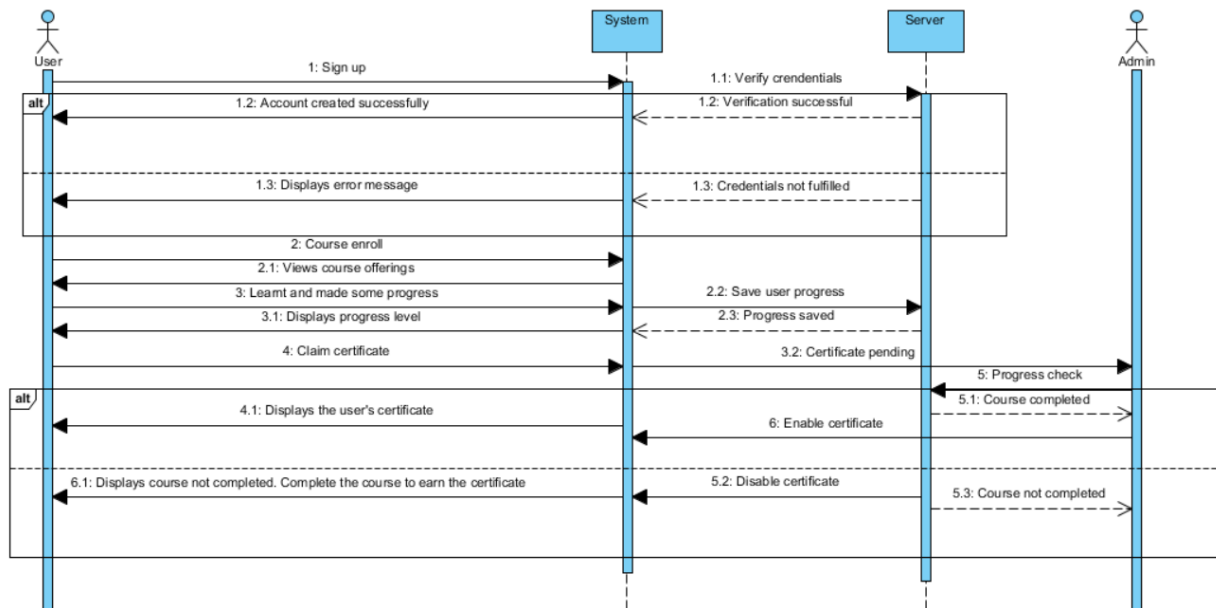


Figure 42: Sequence diagram for user management system

Class diagram

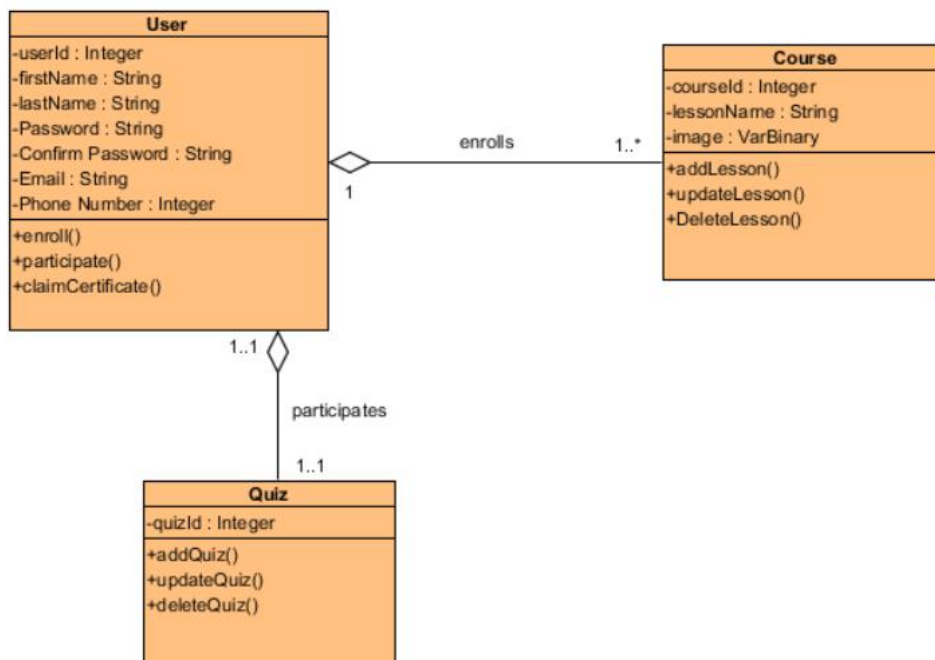


Figure 43: Class diagram for user management system

Bug report

This bug was encountered during the signing up and the error that was shown in the response during the data fetching process from front end to backend was validator error. It was related to the fields that was initialized on the backend for the post purpose.

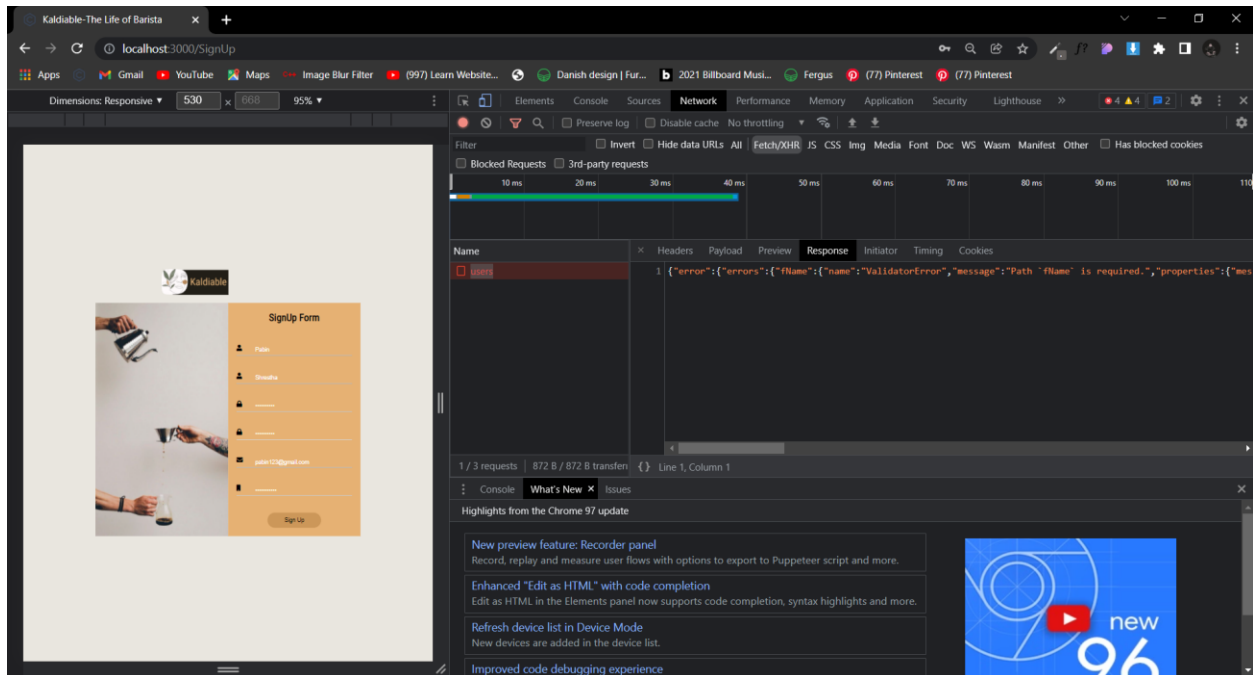
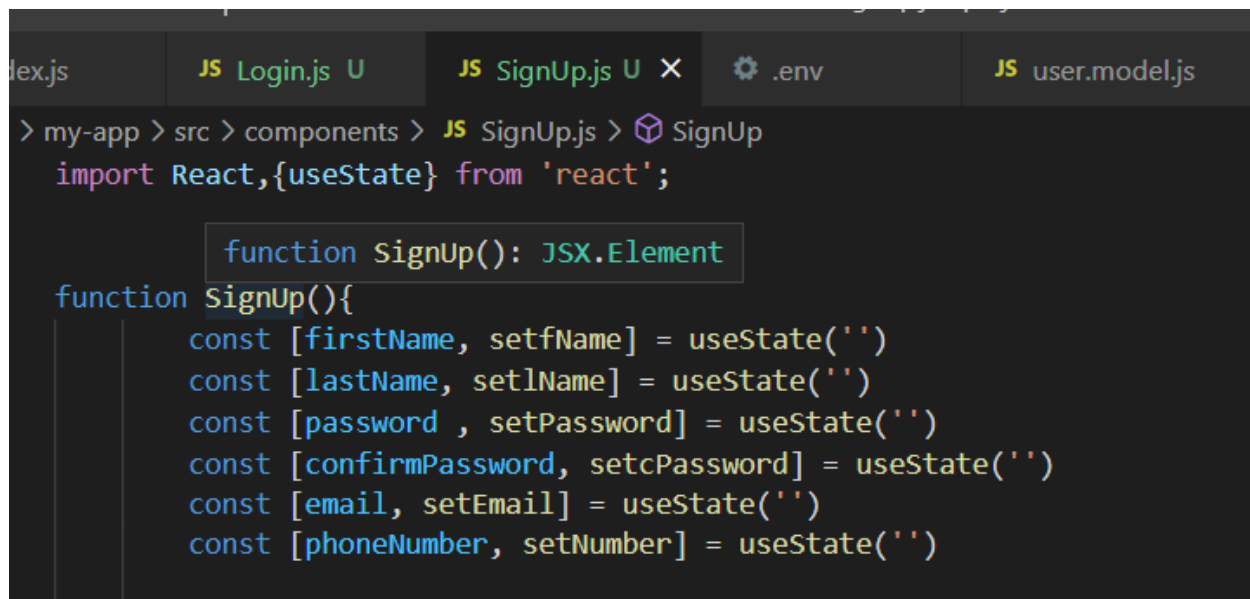


Figure 44: User registration validator error while posting data from front end to backend

The following is the codes provided for fetching data from front end to backend and for each data fetching the field name were initialized as follows.

The use of “useState” add states for functional components it is a hook that that allows to have state variables in that added functional component. The created state can be of any type as the state is an object with hooks. The created states store a single data that can be of integer, array, string, Boolean. (Patel, 2020)

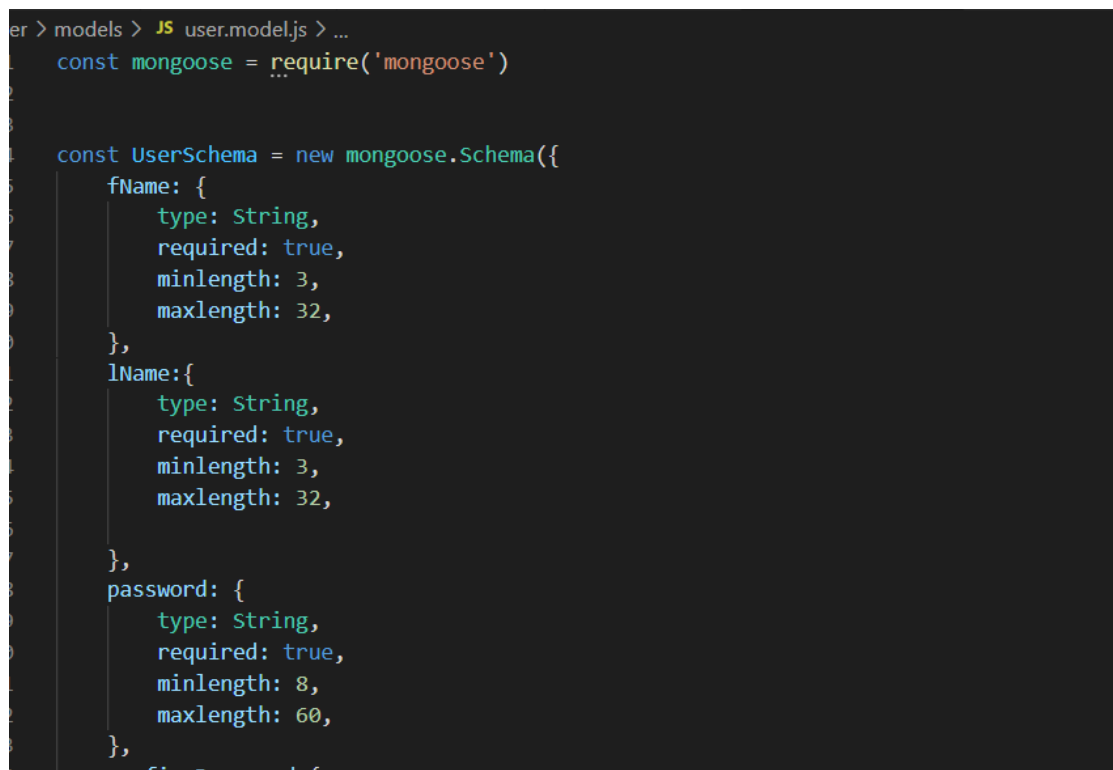


```
JS Login.js U JS SignUp.js U X .env JS user.model.js
> my-app > src > components > JS SignUp.js > SignUp
import React,{useState} from 'react';

function SignUp(): JSX.Element
function SignUp(){
  const [firstName, setfName] = useState('')
  const [lastName, setlName] = useState('')
  const [password , setPassword] = useState('')
  const [confirmPassword, setcPassword] = useState('')
  const [email, setEmail] = useState('')
  const [phoneNumber, setNumber] = useState('')
```

Figure 45: Codes for the fetching of values

The following codes is for the user schema on how the database is going to store the data. The used database for storing of data is MongoDB atlas. Importing mongoose is necessary as it acts as a schema validator for MongoDB.



```
er > models > JS user.model.js > ...
const mongoose = require('mongoose')

const UserSchema = new mongoose.Schema({
  fName: {
    type: String,
    required: true,
    minlength: 3,
    maxlength: 32,
  },
  lName:{
    type: String,
    required: true,
    minlength: 3,
    maxlength: 32,
  },
  password: {
    type: String,
    required: true,
    minlength: 8,
    maxlength: 60,
  },
  confirmPassword: {
```

Figure 46: User Schema

The error that was displaying in the front end got solved when the name of the fields initialized in the front end matched with the fields initialized in the backend.

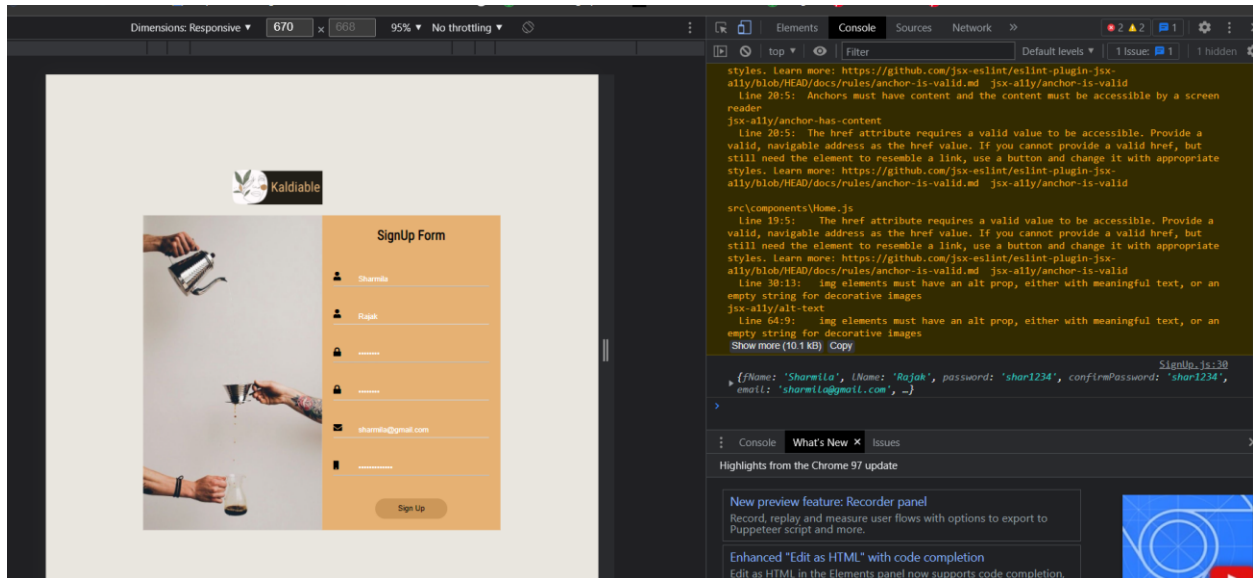


Figure 47: Data fetching successful

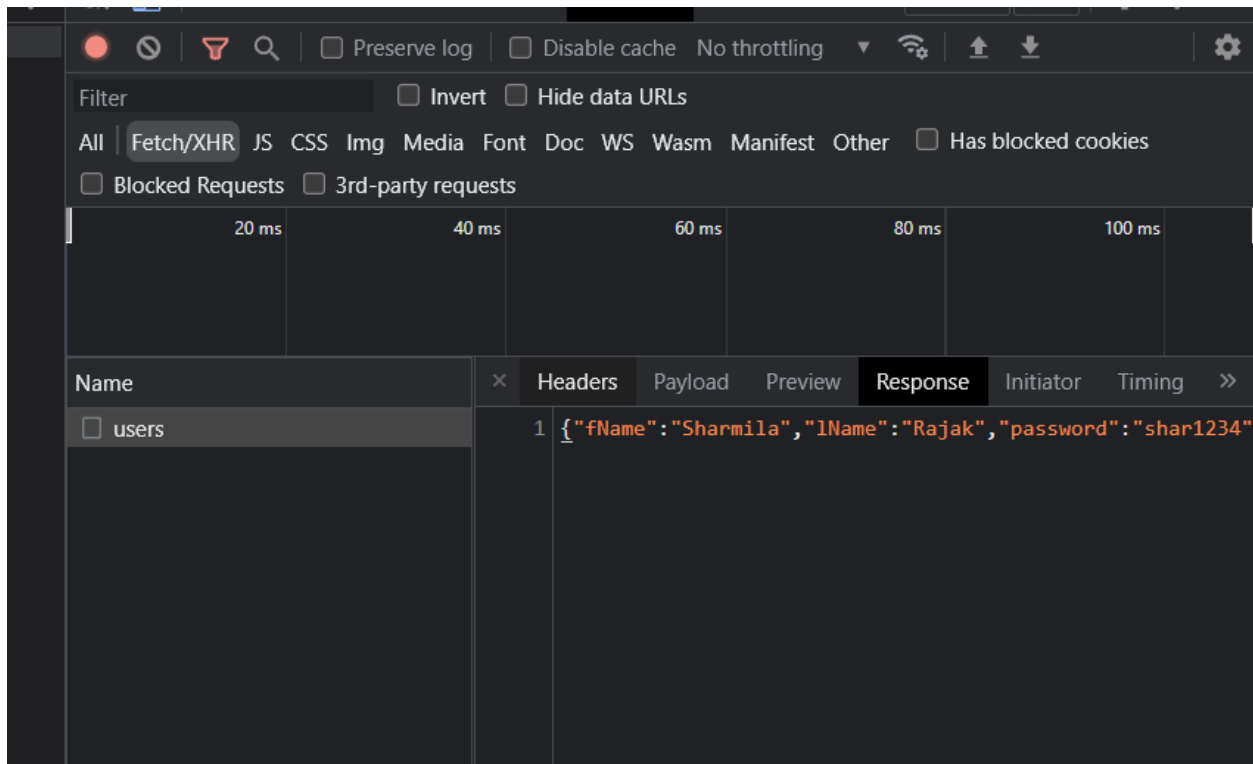


Figure 48: Response fetched in the backend

The code modifications that was done to solve the validator error. This makes it clear that to fetch data from front end to back the fields name should be the same for the proper validation.

```
function SignUp(){
  const [fName, setfName] = useState('')
  const [lName, setlName] = useState('')
  const [password , setPassword] = useState('')
  const [confirmPassword, setcPassword] = useState('')
  const [email, setEmail] = useState('')
  const [phoneNumber, setNumber] = useState('')

  async function registerUser(event){
    event.preventDefault()
    const response = await fetch('http://localhost:3000/api/v1/users', {
      method : 'POST',
      headers: {
        'Content-Type': 'application/json',
      },
      body: JSON.stringify({
        fName,
        lName,
        password,
        confirmPassword,
        email,
        phoneNumber
      })
    })
  }
}
```

Figure 49: Code modification for the proper validation

The fetching of the data from front end and the response of the fetching to of that data in the API. Postman is used for the API testing it allows to check all the request made by the user such as: - post request, get request. The use of postman makes it easier to track errors in the API during any request made.

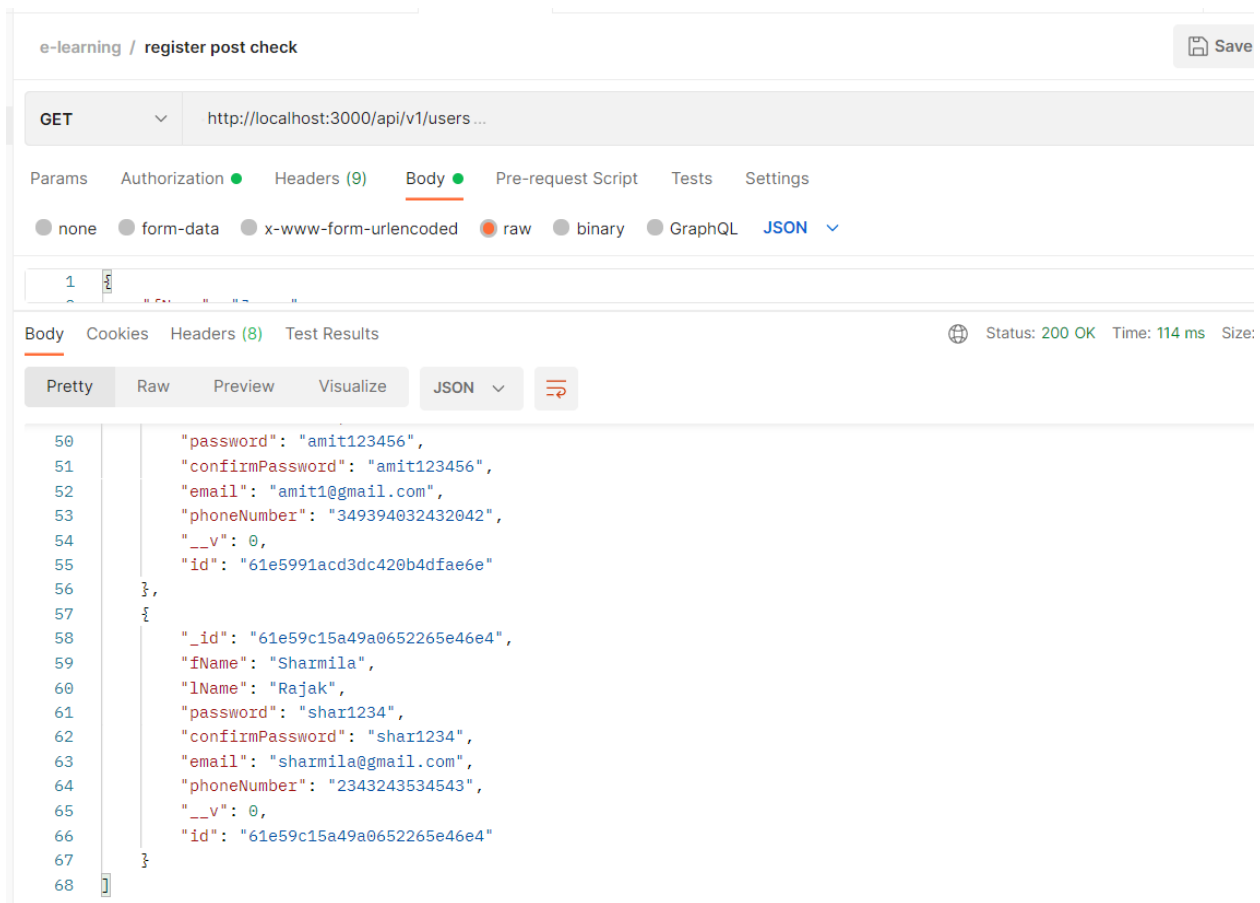


Figure 50: Get request made in Postman

As from the above API it is seen how data is stored in the database. It shows security issue as the password is visible to all when the API is online and request is made. It can cause serious issue if the database of this project is hacked leaving all the user's information clear to the hacker. So, for strong security purpose the password should be in a form that no hacker can understand so password hashing comes handy.

The NPM provides us with a bcrypt package that allows comparing and hashing password that the user submits while accessing the system. The bcrypt library hashes the submitted password and stores that hashed password that cannot be encrypted in its original form creating a strong security measure. Furthermore, when the same user tries to login to the system their inputted password is compared with the hashed password that was previously stored in the database and that comparing functionality is also provided by bcrypt library. (Sebastian, 2021)

```

const bcrypt = require('bcryptjs');

router.get('/', async (req, res) =>{
  const userList = await User.find();
  if(!userList){
    res.status(500).json({success:false})
  }
  res.send(userList);
})

router.post('/',(req,res) =>{
  const user = new User({
    fName: req.body.fName,
    lName: req.body.lName,
    password: bcrypt.hashSync(req.body.password, 10),
    confirmPassword: bcrypt.hashSync(req.body.confirmPassword, 10),
    email: req.body.email,
    phoneNumber: req.body.phoneNumber
  })
})

```

Figure 51: Importing bcrypt library

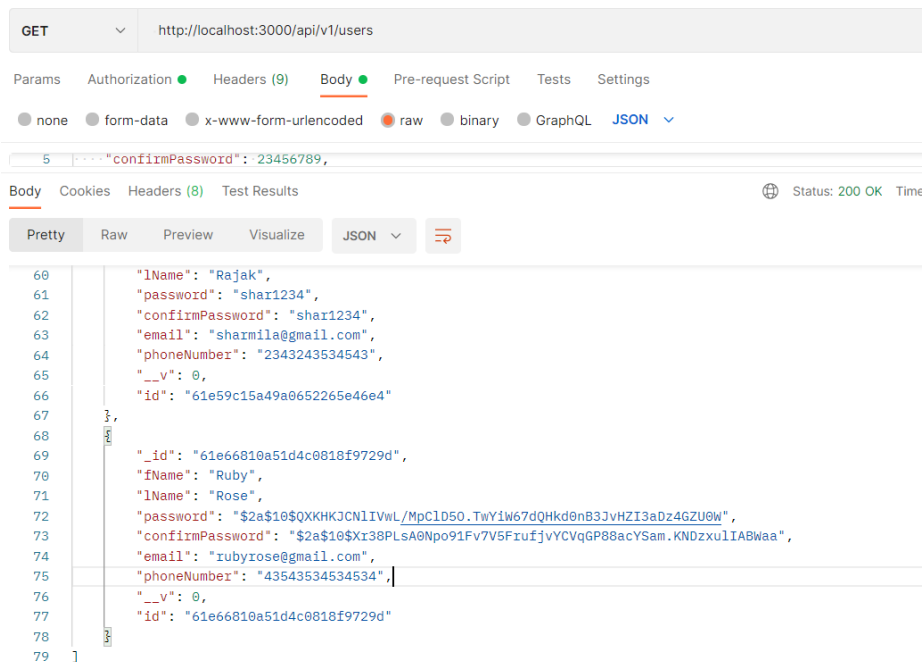


Figure 52: API testing to check the hashing of password



Figure 53: Data visible in the database with hashed password

Suitable try catch was implemented to handle errors that users often make while registering. These line of codes checks whether the entered password is equal to the confirmed password if it is not then a suitable error is displayed on the console.

```
//  
try{  
  if(user.password==user.confirmPassword){  
    user.password = bcrypt.hashSync(req.body.password, 10)  
    user.confirmPassword = bcrypt.hashSync(req.body.confirmPassword, 10)  
    user = await user.save();  
    res.json({message: 'user created', user})  
  }else{  
    return res.json('Password mismatched');  
  }  
}catch(err){  
  res.json(err.message)  
  console.log(err)  
}
```

Figure 54: Code implementation for comparing passwords

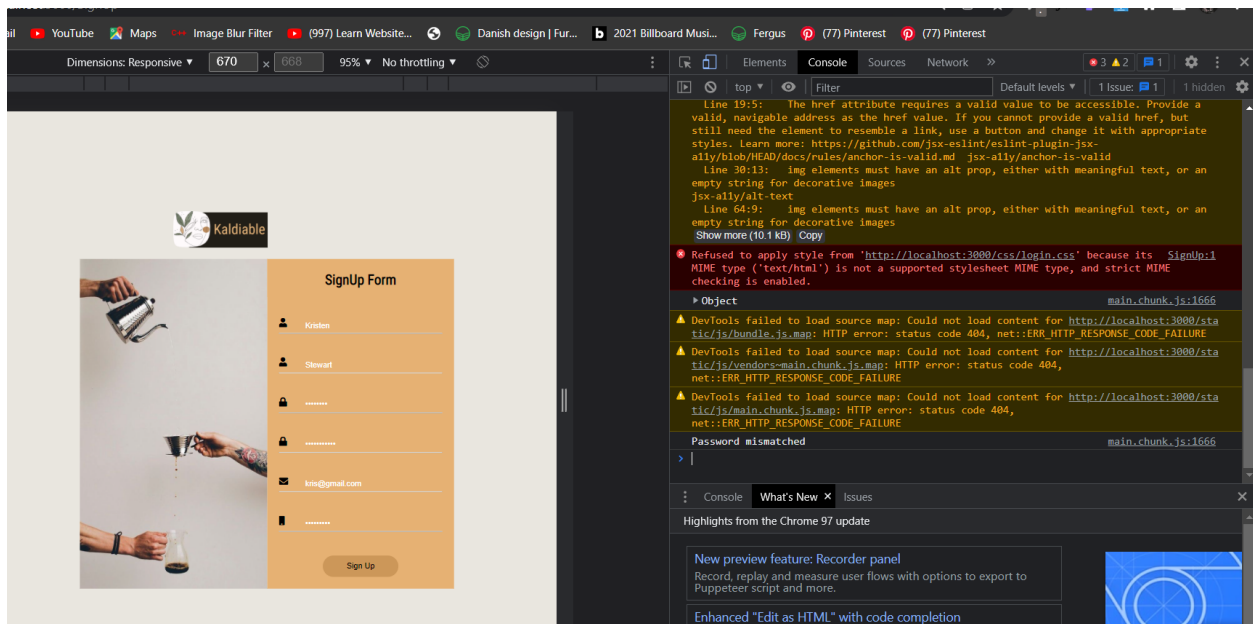
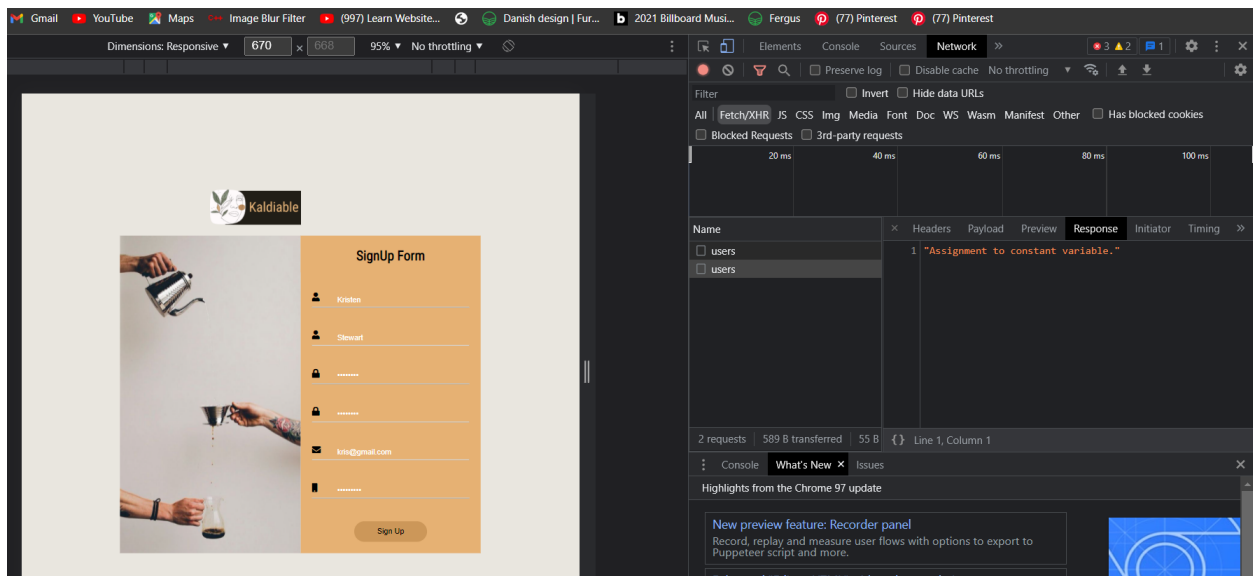


Figure 55: Testing the try catch block on the actual system

After all the credentials of the user registration is met the user gets registered and their information is stored successfully in the database.



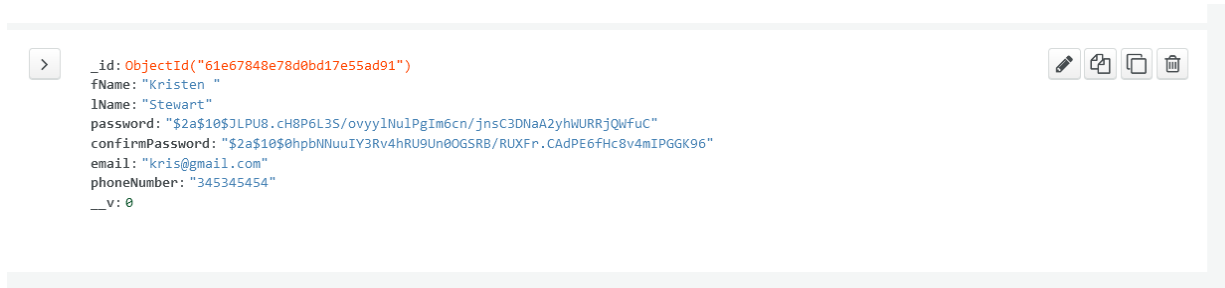


Figure 56: Successful user registration

Course Management System

Req. Code	Req. Description	Use case	Moscow Prioritization
CMS-F-1.0	The admin must be able to add lessons for the course on the system. [create lesson]	Create lesson	Must have
CMS-NF-1.1	The users must be able to view the new created lesson on the system.		
CMS-UR-1.1	The image uploading must be made easier for the admin through simple drag and drop.		
CMS-F-2.0	The system should allow the admin to update the image and description of a provided lesson. [update lesson]	Update Lesson	Should have
CMS-NF-1.1	The description that must be added on the lesson must be able to store appropriate size of words.		
CMS-UR-1.1	The image that needs to be uploaded must be of an appropriate size according to the user interface.		
CMS-F-3.0	The admin must be able to delete the created lesson content provided in the course. [delete lesson]	Delete lesson	could have

Activity diagram

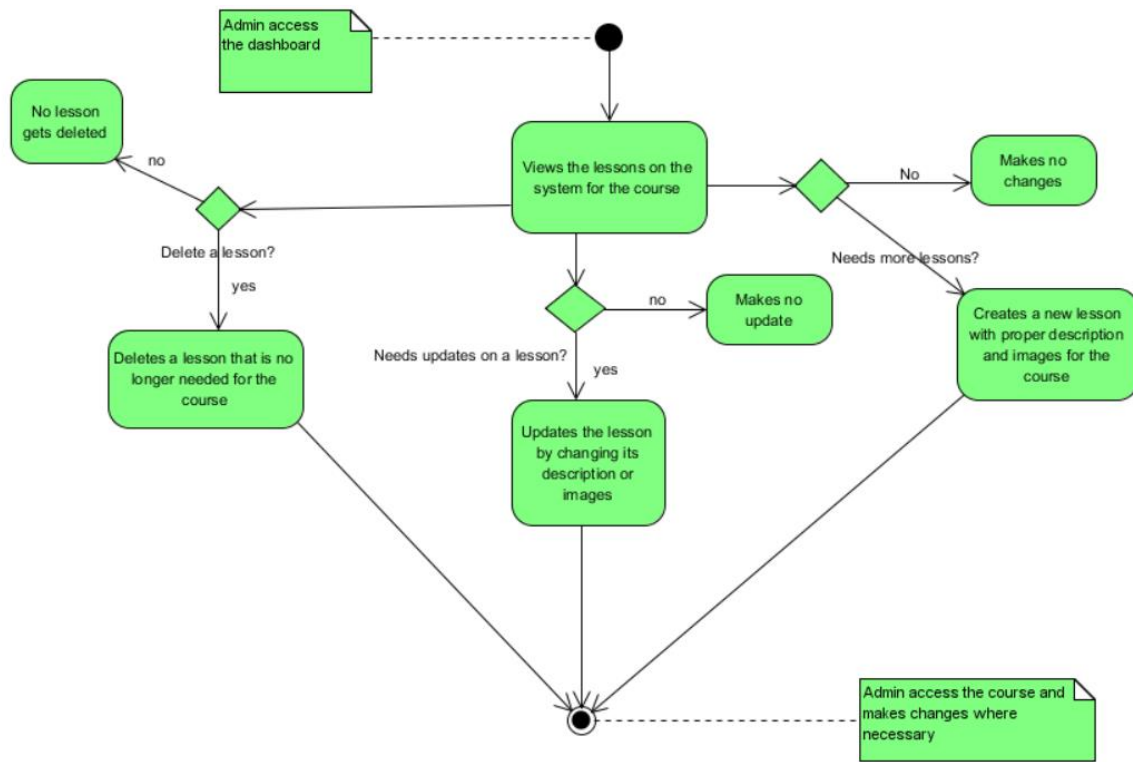


Figure 57: Activity diagram for course management system

Use case diagram

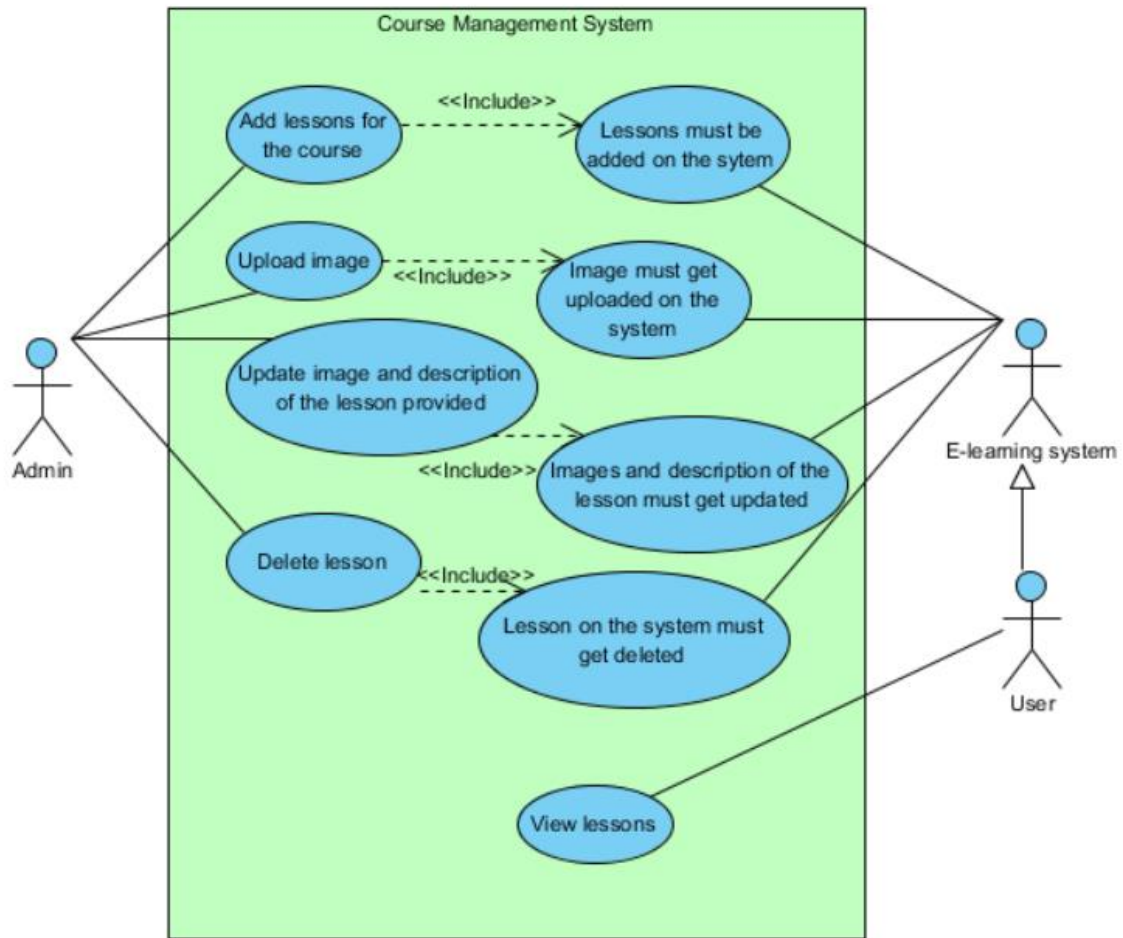


Figure 58: Use case for course management system

Sequence Diagram

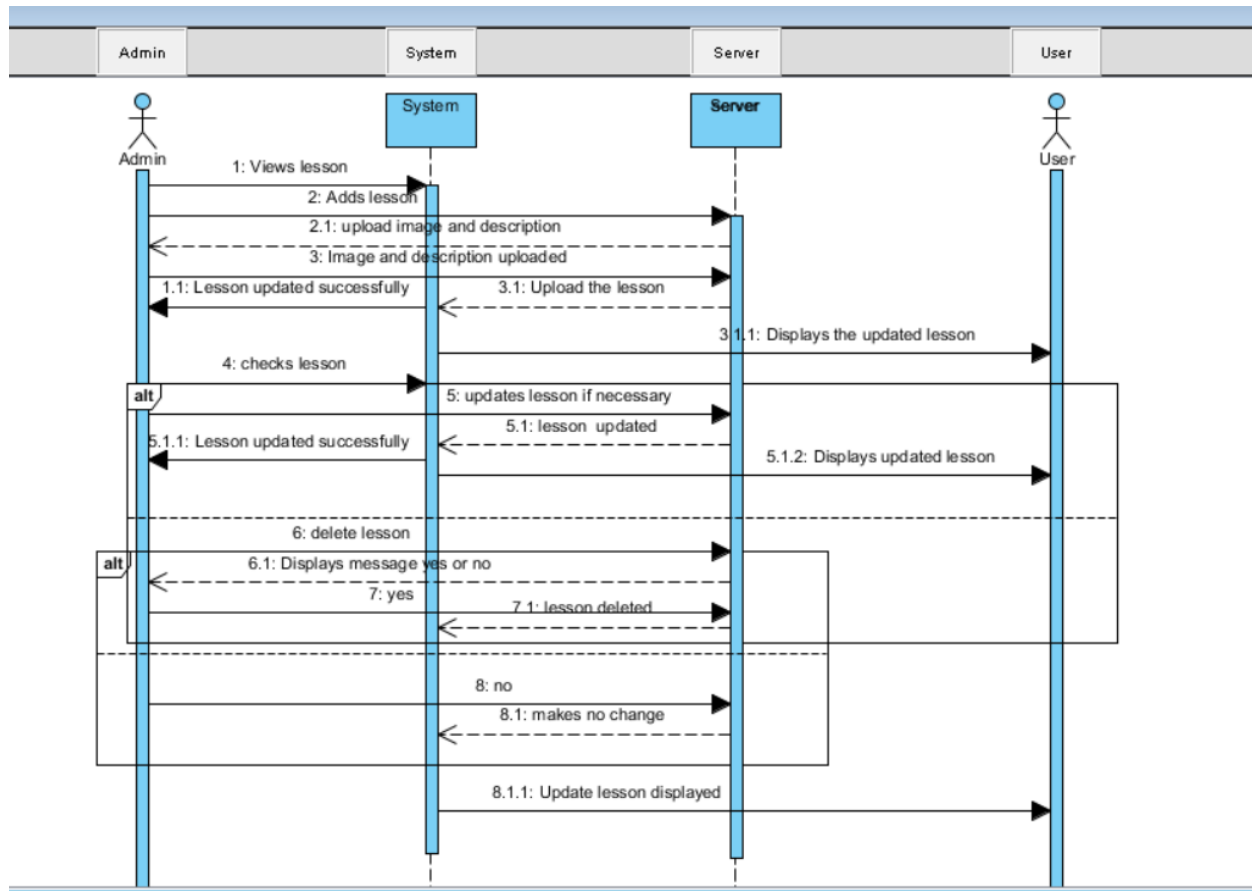


Figure 59: Sequence diagram for course management system

Quiz Management System

Req. Code	Req. Description	Use case	Moscow Prioritization
QMS-F-1.0	The admin must be able to add quiz games with questions and answers on the system. [create quiz games]	Create quiz games	Must have
QMS-NF-1.1	The users must be able to view the quiz games.		
QMS-F-2.0	The users must be able to participate in the quiz game. [Quiz game participation]	Quiz game participation	Should have
QMS-NF-2.1	The users must be able to view the questions and answers for the quiz game.		
QMS-UR-2.1	When the users play the quiz game on answering the correct or incorrect answer the users must be made aware their answers are correct or incorrect through some message or coloration on the user interface.		
QMS-F-3.0	The progress on the games so far achieved by the user must be saved. [Save progress]	Save progress	Could have
QMS-NF-3.1	The users must be able to view their progress on playing the game.		
QMS-UR-3.1	The system must send proper message to let the user know the game is completed.		
QMS-F-4.0	The admin must be able to update the quiz questions and answers. [update questions and answers for the game]	Update questions and answers	
QMS-F-5.0	Admin must be able to use delete the questions and answers for the game. [delete questions and answers for the game]	Delete questions and answers	

Activity diagram

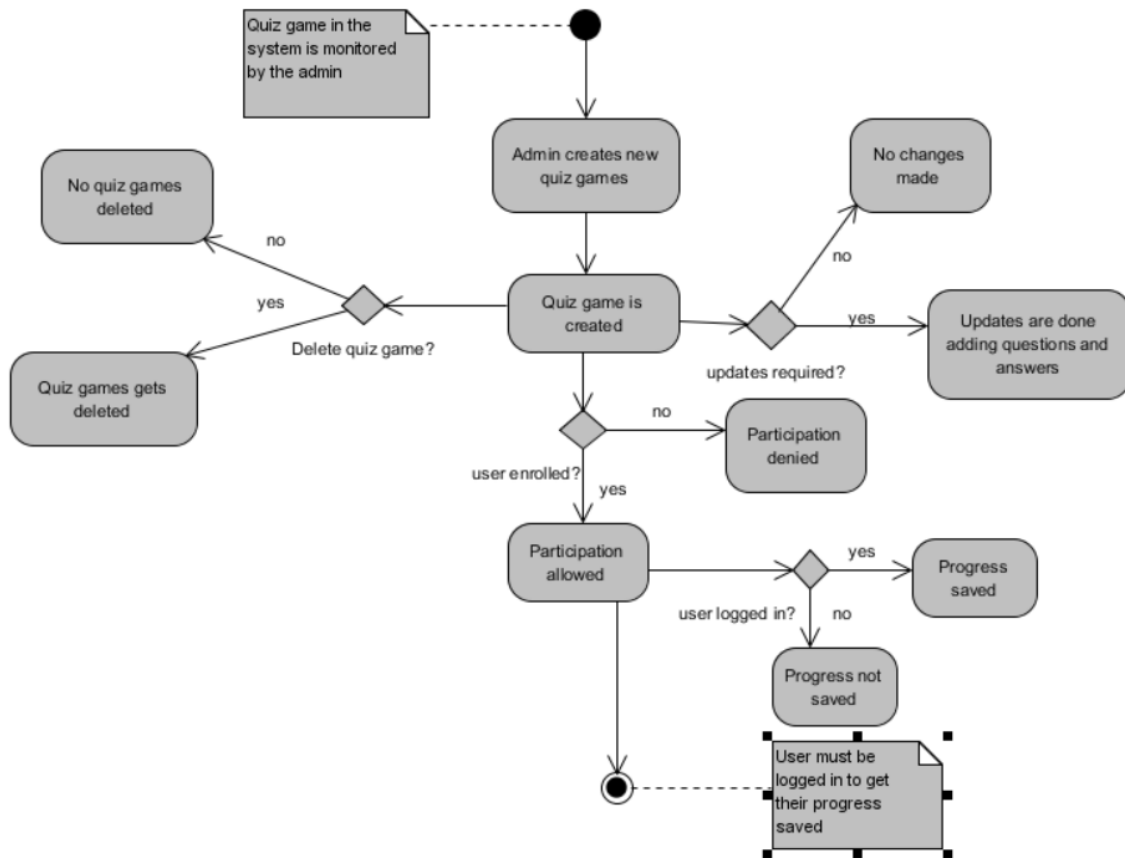


Figure 60: Activity diagram for quiz management system

Use case diagram

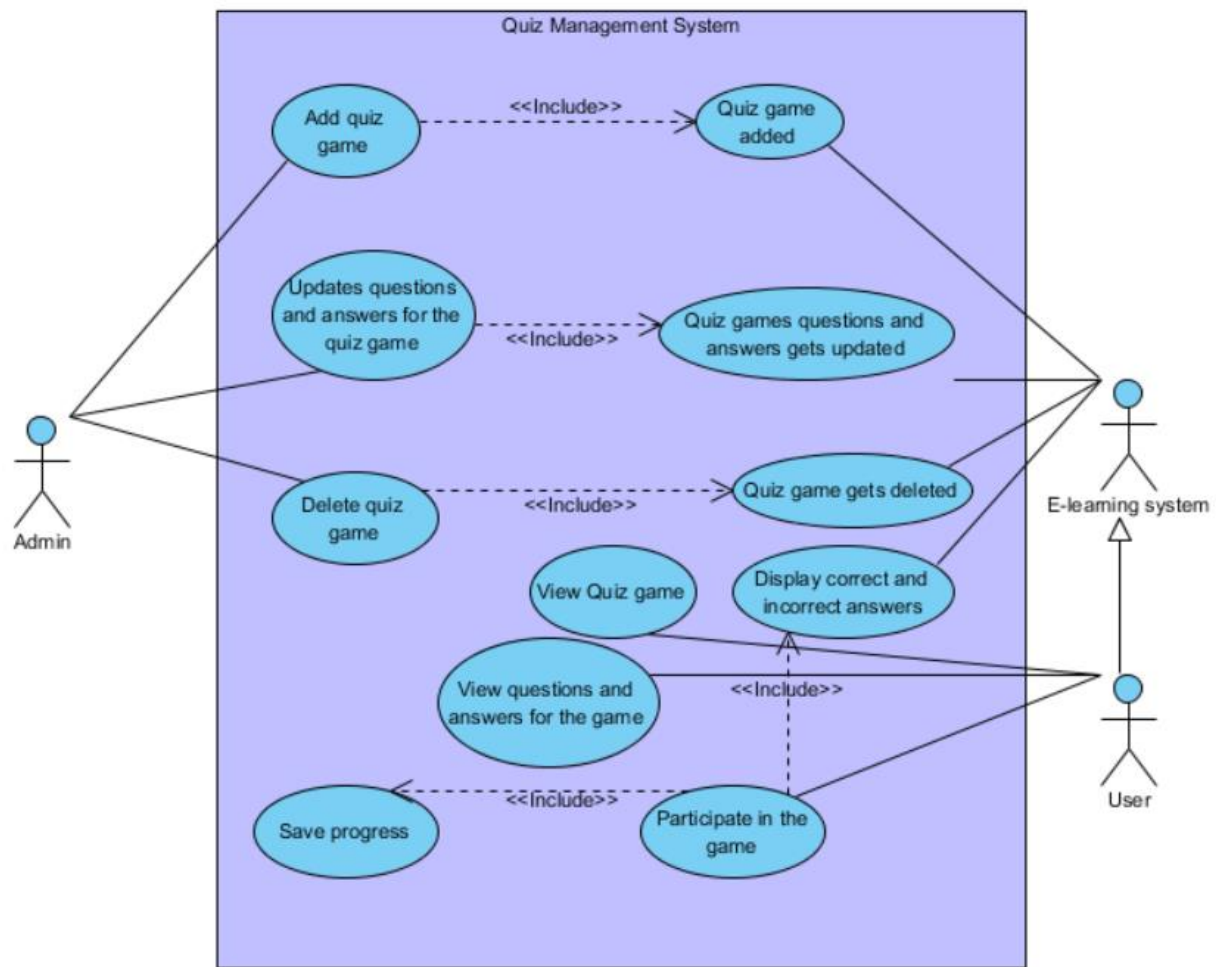


Figure 61: Use case diagram for quiz management system

Sequence diagram

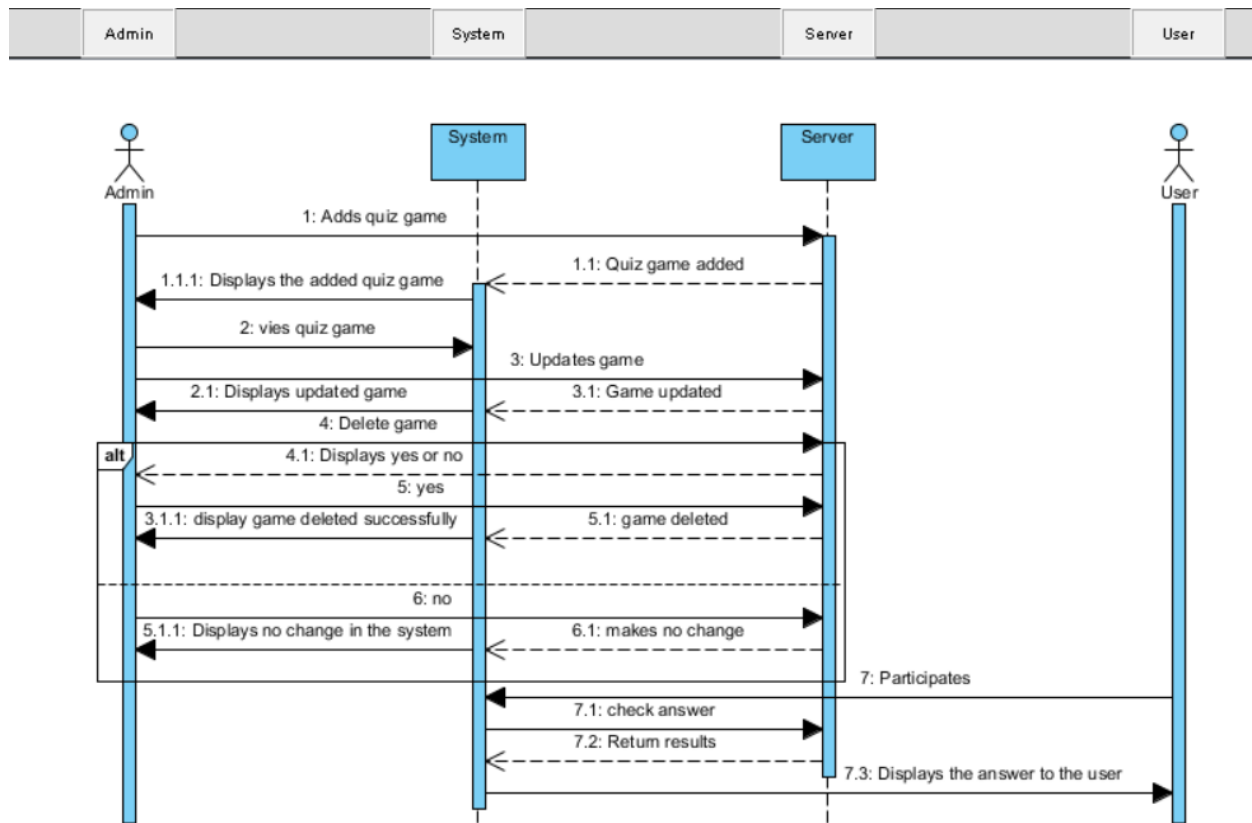


Figure 62: Sequence diagram for quiz management system

Video Management System

Req. Code	Req. Description	Use case	Moscow Prioritization
VMS-F-1.0	The admin must be able to add videos for the course on the system. [add lesson]	Add videos	Must have
VMS-NF-1.1	The users must be able to view and play the video on the system.		
CMS-UR-1.1	The uploaded video must be responsive on every device.		

VMS-F-2.0	The system should allow the admin to delete the video from the system. [delete video]	Delete videos	Could have
-----------	---	---------------	------------

Activity diagram

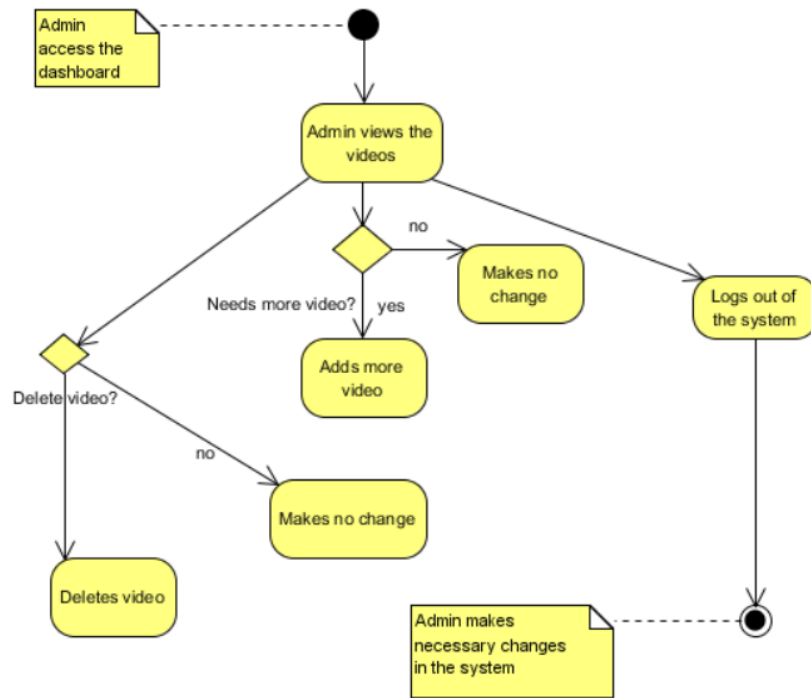


Figure 63: Activity diagram for video management system

Use case diagram

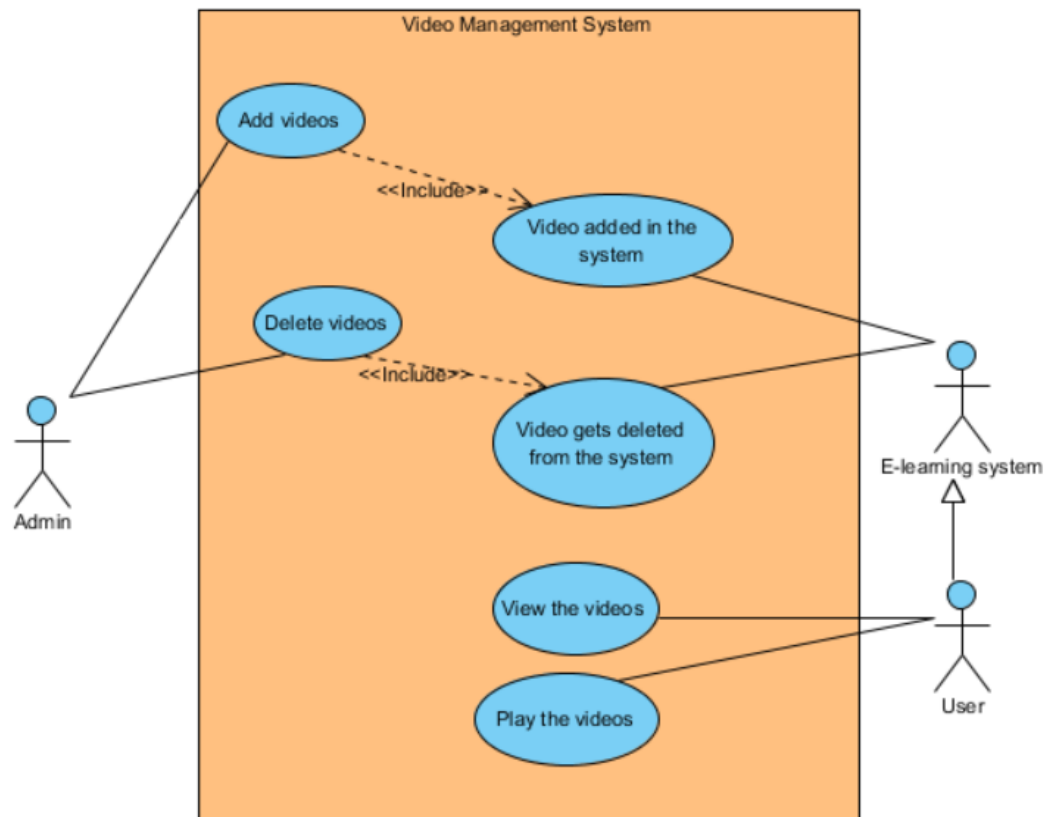


Figure 64: Use case diagram for video management system

Sequence diagram

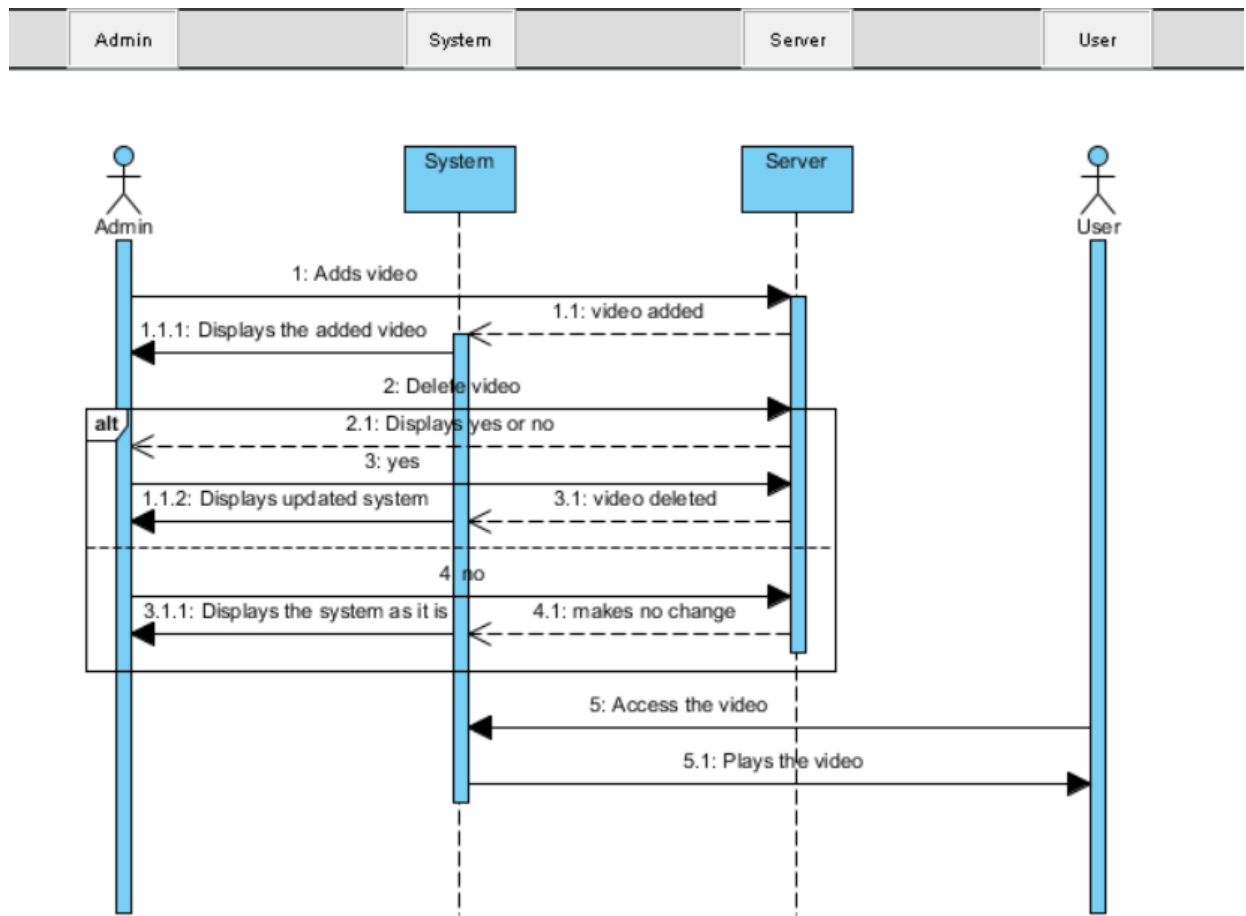


Figure 65: Sequence diagram for video management system

Tools Management System

Req. Code	Req. Description	Use case	Moscow Prioritization
TMS-F-1.0	The admin must be able to create tools necessary for the course.	Create tools	Must have
TMS-NF-1.1	The users must be able to view the tools created by the admin.		
TMS-F-2.0	The users must be able to purchase the tools.	Purchase tools	Could have
TMS-UR-1.1	The uploaded image must be made easier by allowing the drag and drop feature.		
TMS-F-3.0	The system should allow the admin to delete the tools. [delete tools]	Delete tools	Should have

Activity diagram

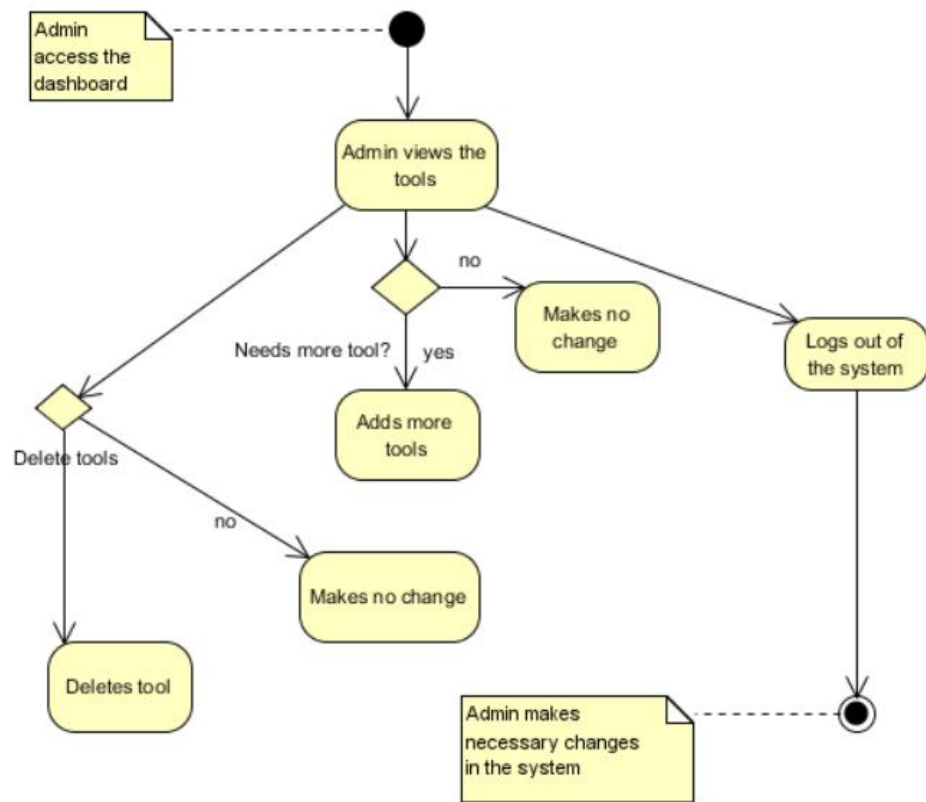


Figure 66: Activity diagram for tools management system

Use case diagram

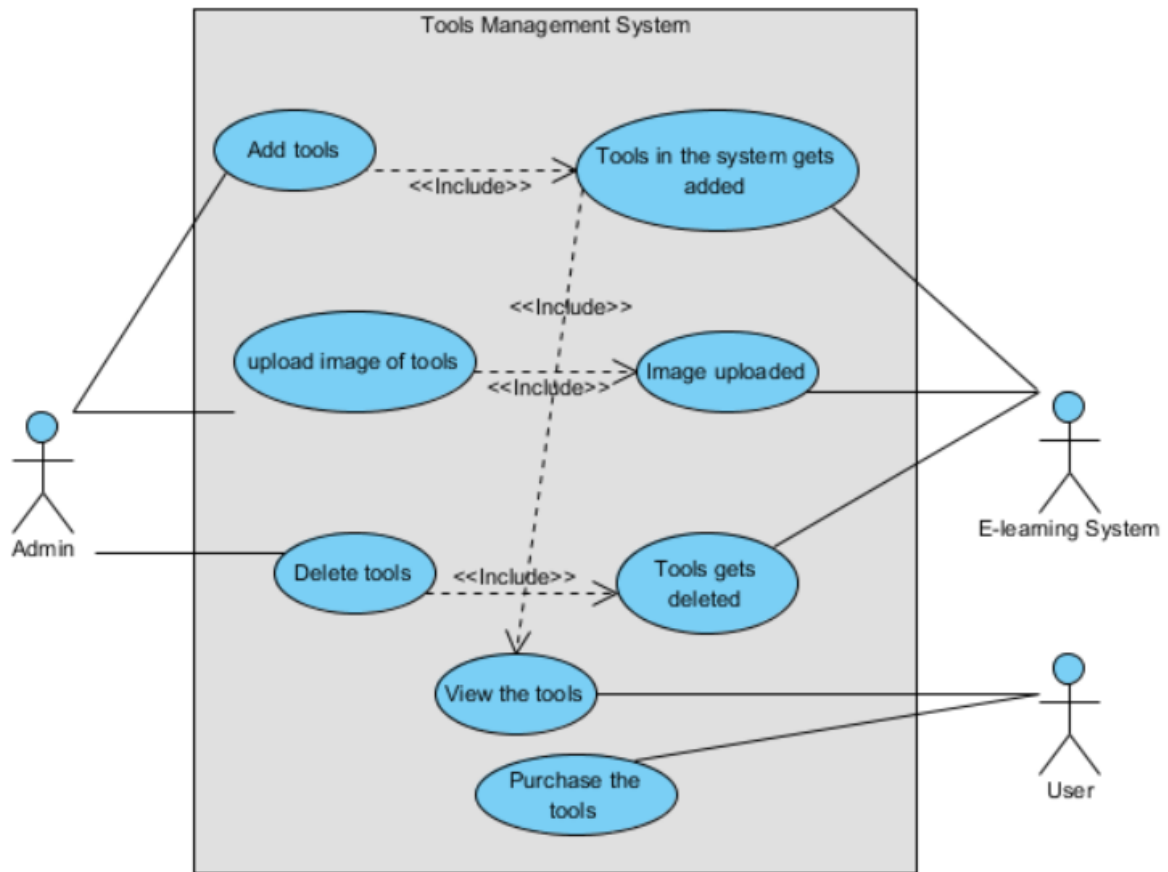


Figure 67: Use case diagram for tools management system

Sequence diagram

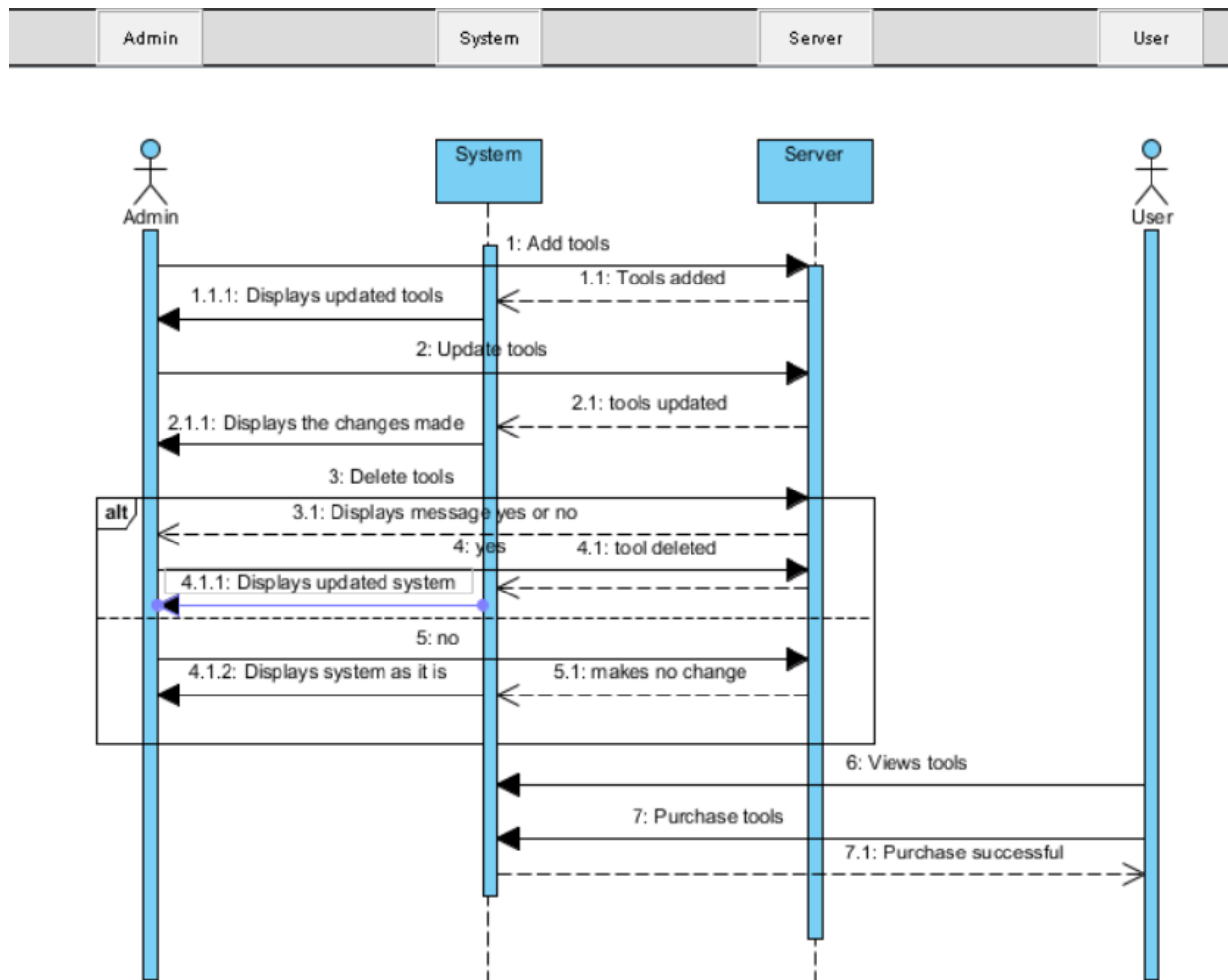


Figure 68: Sequence diagram for tools management system

Class diagram

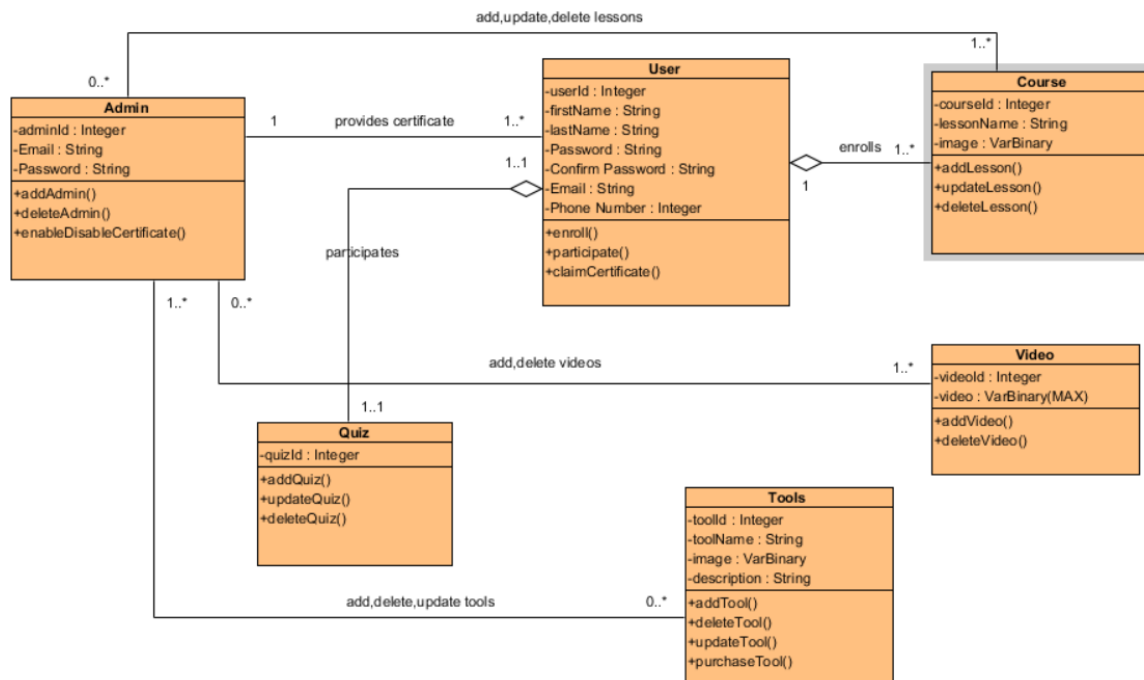


Figure 69: Class diagram for the system

Entity Relationship diagram

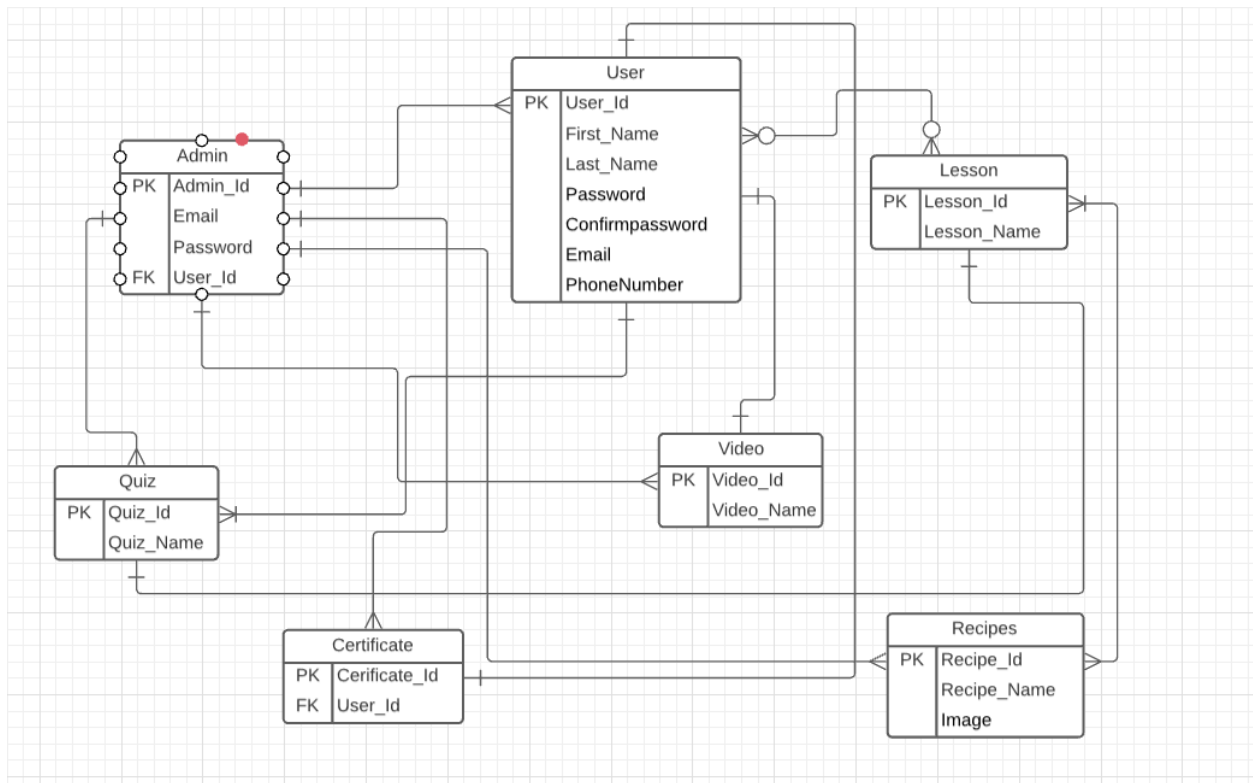


Figure 70: Entity Relationship diagram for System

Wireframe

Logo

Image

SignUp Form

First Name

Last Name

Password

Confirm Password

Email

Phone number

Button

Logo

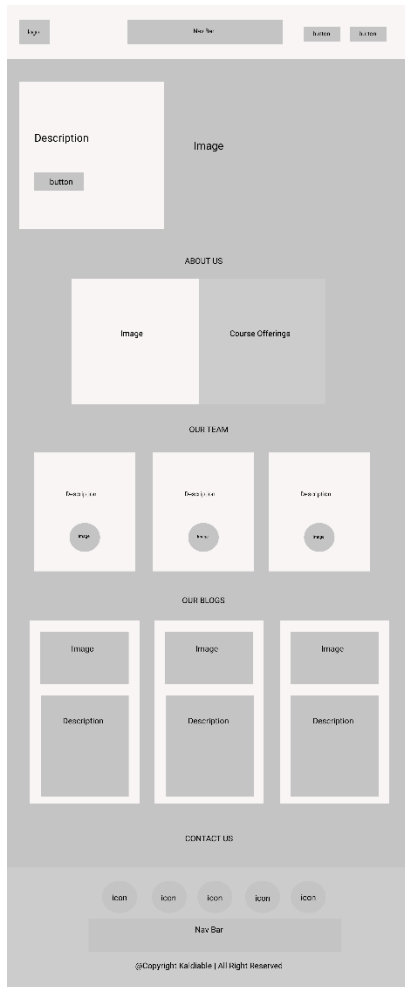
Image

Login Form

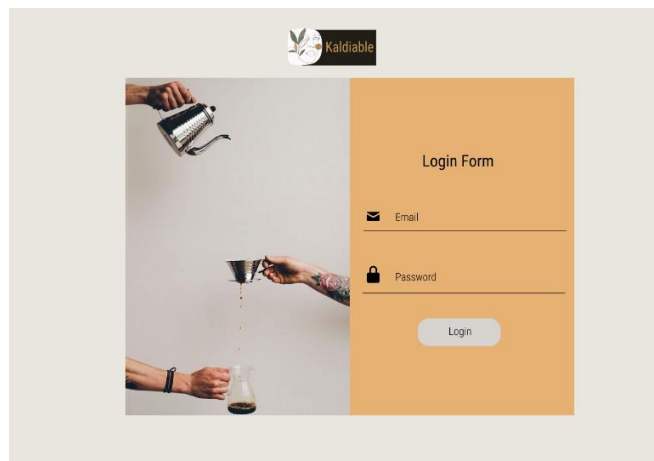
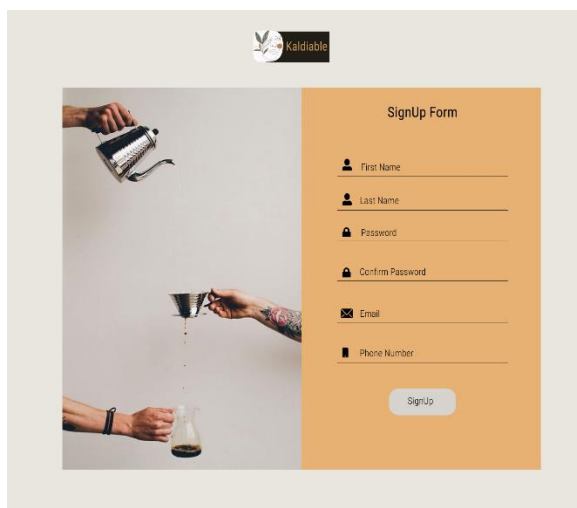
Email

Password

Button



UI UX designs for the system



12. Log sheet

Faculty of Science and Engineering
School of Mathematics and Computer Science



PROJECT MANAGEMENT LOG	
First Name: Jeshmi	Surname: Rajak
Student Number: 2049857	Supervisor: Santosh Sapkota
Project Title: Kaldiable	Month: November
What have you done since last meeting	
Features are finalized, artefact of project is designed, proposal of the project is reviewed	
What do you aim to complete before the next meeting	
Wireframes of the web pages	
Supervisor Comments	
- Description must be provided of the gantt chart - Wireframes should be added - Provide details of testing in the Artefact section	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: Jeshmi Rajak

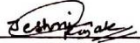
Date: 11th Nov 2021

Supervisor Signature: [Signature]

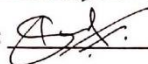
Date: 14th Nov 2021

PROJECT MANAGEMENT LOG	
First Name: Jeshmi	Surname: Rajak
Student Number: 2049857	Supervisor: Santosh Sapkota
Project Title: Kaldiable (E-learning)	Month: December
What have you done since last meeting	
Literature review for the similar system, user authentication	
What do you aim to complete before the next meeting	
Designs, OTP authentication, user registration and login development	
Supervisor Comments	
Work on the front end development and learn more about react	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 12/12/2021

Supervisor Signature: 

Date: 12/12/2021

PROJECT MANAGEMENT LOG	
First Name: JESMI	Surname: RAJAN
Student Number: 2049857	Supervisor: Santosh Sapkota
Project Title: Kaldiable (E-learning)	Month: December
What have you done since last meeting	
Front-end development with designs and wireframes	
What do you aim to complete before the next meeting	
Front-end development that is developed using HTML CSS and Javascript to react and SRS of User Management system	
Supervisor Comments	
Admin panel template, sprint backlog	

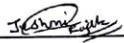
We confirm that the information given in this form is true, complete and accurate.

Student Signature:  Date: 19/12/2021

Supervisor Signature:  Date: 19/12/2021

PROJECT MANAGEMENT LOG	
First Name: Jeshmi	Surname: Rajak
Student Number: 2649857	Supervisor: Santosh Sapkota
Project Title: Kaldiable	Month: January
What have you done since last meeting	
Front-end development from html, css to React, admin panel	
What do you aim to complete before the next meeting	
Back-end for user management system along with the front end, SRS for user management system, FDD	
Supervisor Comments	
Work on backend Development.	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

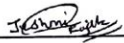
Date: 9/01/2022

Supervisor Signature: 

Date: 3/02/2022

PROJECT MANAGEMENT LOG	
First Name: Jeshmi	Surname: Rajak
Student Number: 2649857	Supervisor: Santosh Sapkota
Project Title: Kaldiable	Month: January
What have you done since last meeting	
Front-end development from html, css to React, admin panel	
What do you aim to complete before the next meeting	
Back-end for user management system along with the front end, SRS for user management system, FDD	
Supervisor Comments	
Work on backend Development.	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 9/01/2022

Supervisor Signature: 

Date: 3/02/2022

PROJECT MANAGEMENT LOG	
First Name: JESHMA	Surname: RAJAK
Student Number: 2049852	Supervisor: Santosh Sapkota
Project Title: Kaldiable (E-learning)	Month: February
What have you done since last meeting	
Main pages designs and development	
What do you aim to complete before the next meeting	
Research on all the four aspects, laws and prepare a report.	
Supervisor Comments	
Include wireframes on the SRS artifact, git push the development done so far	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 05/02/2022

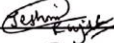
Supervisor Signature: 

Date: 05/02/2022

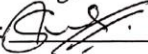
Faculty of Science and Engineering
School of Mathematics and Computer Science

PROJECT MANAGEMENT LOG	
First Name: JESMI	Surname: RAJAK
Student Number: 2049857	Supervisor: Santosh Sapkota
Project Title: Kaldiable (E-learning)	Month: January
What have you done since the last meeting	
Worked on the backend of user management system and successfully joined the front end of sign up and login page with the backend SRS for the sub system for the project. Bug report, Test case for user management system	
What do you aim to complete before the next meeting	
Complete artifact with all the necessary diagrams. Also user roles for the system.	
Supervisor comments	
Work on the admin panel	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 1/22/2022

Supervisor Signature: 

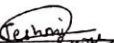
Date: 1/22/2022

Faculty of Science and Engineering
School of Mathematics and Computer Science



PROJECT MANAGEMENT LOG	
First Name: JESMI	Surname: RAJAK
Student Number: 2099857	Supervisor: Santosh Sapkota
Project Title: Kaldiable	Month: March
What have you done since the last meeting	
Email Verification through OTP authentication and reset password through token.	
What do you aim to complete before the next meeting	
Front end connection with the backend of email verification after user sign in.	
Supervisor comments	
Make a use case diagram for authentication.	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

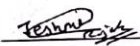
Date: 31/4/2022

Supervisor Signature: 

Date: 31/4/2022

PROJECT MANAGEMENT LOG	
First Name: JESMI	Surname: RAJAK
Student Number: 2049857	Supervisor: Santosh Sapkota
Project Title: Kaldialle	Month: March
What have you done since the last meeting	
Front end connection with the backend with reset password	
What do you aim to complete before the next meeting	
Admin panel with user role assigned to the specific users	
Supervisor comments	
Work on the front-end of the category page	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 18th March 2022

Supervisor Signature: 

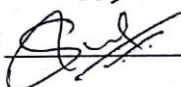
Date: 18th March 2022

PROJECT MANAGEMENT LOG	
First Name: JESMI	Surname: RAJAK
Student Number: 2049857	Supervisor: Santosh Sapkota
Project Title: Kaldiala	Month: March
What have you done since the last meeting	
Admin panel with crud functionality with user roles.	
What do you aim to complete before the next meeting	
Further backend development with crud functionality	
Supervisor comments	
Update the report of your project according to the changes made.	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 25th March 2022

Supervisor Signature: 

Date: 25th March 2022

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